Ground Truth

Measurements Lead · Station 12, where lightning is made on purpose

15 DAYS · 52 STOPS · GRADE 12 · GROUNDTRUTH

THE OPENING CARD

Station 12 makes lightning on purpose: a rocket flies a wire up, and the strike comes down it. Last August one strike killed every card in a trailer that touched nothing it struck. You are the measurements lead, and no number leaves here without its working. In fifteen days the review reads the season report and gives this place a second mast or a padlock. Six people walk out under that mast on a rule you have not written yet.

What the flat is for

BEFORE THE DAY — GREET

Station 12 is six people and a storm season

Walk the site once and put a job to every name you meet, because that is who you will be asking for numbers all season.

PLAN CARD 126W

Monday of week three. Three weeks of the season are left. The season report takes one number a day. Each one goes in with the working that turns what the instrument saw into what is claimed. Adeyinka Vero, the station lead, wants shots. Hal Brenner, the safety officer, wants the launch rule written down first. He wants it agreed before the sky is overhead, not while it is. The trailer that lost every card last August stands two hundred metres from the mast. Its earthing had been signed off that spring, and nobody can say what got in. Today you set the launch rule the crew will fire by. Then you read what a field mill is telling you, and what that April certificate actually measured.

BRIEFING 16W

Week three of six, and the first cell of the week is forecast for the afternoon.

WHAT YOU DECIDE 11W

Establish what the station measures and what last season cost it.

1.1 One call, and a cell coming in

LAUNCH & RECORDS · LAUNCH CONTROL · A ROOM · TRIGGER

SCENE 39W

Operations and Safety lead Hal Brenner has the afternoon cell on radar and the four field mills on the wall. Three actions must finish before a rocket leaves the rail, and the cell is already moving toward the flat.

GUIDE 74W

One rule, written before the cell arrives and held to once the board is released. The walkback takes half an hour from the moment it is called, and by the time the field is at strike conditions everybody has to be behind the bunker already. So the line goes back from strike conditions by that half hour — and not so far back that the pad is cleared for a cell which never built.

ASK 17W

At what surface field do you clear the pad and walk the crew back to the bunker?

BEHIND THE BUTTON 140W

Why the field rather than the radar. The mills report the electric field at the flat, which is what actually determines whether a rocket trailing a wire will trigger a strike. Radar tells you a cell is coming; the field tells you the flat is ready. Only one of those is the launch condition.

What lead time means for each stage. Clearing personnel, arming and launching all have durations, and they are not the same. A threshold set for the longest action is too early for the shortest, which is why there are three numbers to write rather than one.

Why the board is signed first. A threshold adjusted while the cell comes in is not a rule; it is a description of what somebody already decided to do. Committing beforehand is what makes the afternoon attributable to a plan.

TAKES AS READ

a crew on an open flat has to be walked back before a strike is likely · a forecast of where a cell will be is not a promise · charge, and where it sits on a conductor — taken as read

VERDICT 75W

A launch rule is a set of promises about time, not about field strength. The field is only the clock face. Each threshold must start an action early enough to finish before the sky becomes dangerous. Setting thresholds late is the natural error. Early calls look wasteful in hindsight, while late calls look acceptable until the one afternoon they fail. Writing the numbers first makes the decision reviewable and separates a criterion from a preference.

TAKEAWAY 18W

A threshold is only useful if the action it triggers can still be finished before the sky arrives.

1.2 A spinning shutter and a sign

FIELD & CHARGE · FIELD STATION · A ROOM · CHOICE

SCENE 40W

Field-instruments specialist Mira Halvorsen, the field instruments technician, has the reference mill open on the bench, its shutter turning over the plate. On the wall, all four mills read about −1.2 kV/m, and fair-weather readings carry the same negative sign.

GUIDE 67W

All four options are things somebody might want from a mill. Ask of each whether one reading at the ground could contain it. The field under a wide charged sheet does not depend on how high the sheet is. So height drops out, and with it anything that needs height or area. Timing needs a history rather than a reading. That is a limit, not a fault.

ASK 11W

What does the mill's reading tell you about the charge overhead?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a field is a vector and has a direction as well as a size

VERDICT 75W

The field under a wide charged sheet does not depend on the sheet's height. A ground reading can therefore fix charge per square metre while leaving height completely open. That is useful, not defective. One instrument near the ground can report a property of a cloud base more than a kilometre up. It still cannot give total charge without the layer's area. It also cannot give timing without the history of how the reading changed.

TAKEAWAY 18W

> A single field measurement constrains only the quantities that enter the field relation; unmeasured geometry can remain free.

1.3 Twenty-five ohms, at what

EARTHING & TRANSIENTS · EARTHING COMPOUND · A PERSON · CHOICE

SCENE 38W

Ana Sifuentes, the earthing engineer, sets the April certificate on the bench: 25 ohms, instrument named, operator signed. She wants the measured quantity stated exactly before anyone uses the number to make a claim about the storm season.

GUIDE 65W

Four readings of the same certificate. Ask of each what the tester actually did: it pushed a small steady current into the soil and measured the voltage. So the number belongs to steady current. A strike changes by tens of kiloamps in a microsecond, which is a different regime. Nothing on the certificate is wrong. The error would be treating it as the whole story.

ASK 6W

What does the 25-ohm figure describe?

BEHIND THE BUTTON 195W

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TAKES AS READ

an instrument measures under the conditions it applies, not the ones you care about · current as a flow, and current density — taken as read · electric potential as a line integral of the field — taken as read

VERDICT 75W

An earth tester injects a small steady current and measures the voltage it produces. Its 25-ohm result is therefore a steady-current resistance. A strike is a different regime. Its current changes by tens of kiloamps in about a microsecond, so other terms can dominate the voltage. Nothing about the April certificate is wrong. The mistake would be using that steady-state number as though it were a complete description of what the grid does during lightning.

> **TAKEAWAY** 22W
>
> A measurement carries the conditions it was made under, and a number quoted without them is a number about nothing in particular.

END OF THE DAY 12W

A criterion written while the cell is overhead is not a criterion.

DAY 2 OF 15

The layer overhead

PLAN CARD 115W

Tuesday. Yesterday's cell left a number nobody has used. At the moment the crew was walked back, the four field mills read 4.5 kV/m. Vero wants that in the season log as a charge, not as a field. A field is what the meter reads. A charge is what the cloud holds. Emil Strand, the high-voltage engineer, says the hall can make the same field between two plates a metre apart. The arithmetic is the same either way. Today you turn a reading taken at head height into the charge on each square metre of cloud base. The same law says something about the inside of the screened room. Strand wants that written down too.

BRIEFING 15W

The mills read 4.5 kV/m under yesterday's cell, and nobody has turned that into charge.

WHAT YOU DECIDE 15W

Get the charge on the cloud base out of a reading taken at the ground.

2.1 A pillbox through the cloud base

FIELD & CHARGE · FIELD STATION · A ROOM · DERIVE

SCENE 37W

Field-instruments specialist Mira Halvorsen has 4.5 kV/m on the board. Station lead Adeyinka Vero wants coulombs per square metre. The charge sits near cloud base, spread over an area far wider than its height above the flat.

GUIDE 66W

You build the working a line at a time. At each step you choose the expression the line above actually gives you. Every wrong branch is legal algebra, so the malformed-line trick will not help. Start from Gauss's law and choose a surface the geometry makes easy. The layer is far wider than it is high, so a pillbox straddling it is the surface that pays.

ASK 15W

Turn the mill reading into the charge per square metre on the layer above it.

BEHIND THE BUTTON 165W

Why a pillbox. Gauss's law is true for any closed surface and useful only for one that matches the symmetry. A charged layer much wider than its height looks locally like an infinite sheet. A short flat box through it therefore has field passing through one face and nothing through the sides.

Where the factor of two goes. An isolated sheet drives field out of both faces, giving $E = \sigma/2\varepsilon_0$. A layer above ground with a conducting flat beneath it has the field on one side only, so the same charge gives twice the field. Which case you are in is a physical question, not an algebraic one.

Why Vero wants coulombs per square metre. A field in kilovolts per metre is what the mill reports. A charge density is what can be compared with other cells, other days and the literature. Converting the reading into the quantity that caused it is the whole point of having a model of the cloud at all.

TAKES AS READ

flux out of a closed surface counts the charge inside it · a layer far wider than its distance can be treated as unbounded

VERDICT 78W

The result is 8.0×10^{-8} C/m². Over ten square kilometres, that corresponds to about 0.8 C on the layer. The derivation matters more than the total. Choose a pillbox so E is constant on its two faces and parallel to its sides. Then E can leave the surface integral. The cloud-base height never appears. That is why a mill at the ground can recover charge per unit area on a layer more than a kilometre above it.

TAKEAWAY 24W

A wide sheet of charge produces a field that does not depend on distance, which is why a reading at the ground fixes it.

2.2 Two plates, and a factor of two

IMPULSE HALL · A PERSON · CHOICE

SCENE 41W

High-voltage engineer Strand has two plates a metre apart with equal and opposite surface charge. Beside them is the one-sheet model from yesterday. The two setups use the same σ, and he asks why the field between the plates is different.

GUIDE 60W

Four explanations for a factor of two. Ask of each whether it changes the law or adds a source. Both setups use the same charge per unit area, so nothing about the charge has doubled. What has changed is how many sheets contribute in the gap. And note what happens outside the pair, where the two contributions point opposite ways.

ASK 14W

Why is the field between two charged plates twice that of one charged layer?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

fields from separate charges add as vectors

VERDICT 79W

One charged sheet gives $\sigma/(2\varepsilon_0)$ on each side. Put an oppositely charged sheet across the gap and its field points the same way between the plates. The two contributions therefore add to σ/ε_0. Outside the pair, the fields point in opposite directions and cancel. That is why an ideal capacitor keeps most of its field between the plates. Gauss's law has not changed. The factor of two comes from superposition: two sources contribute to the field in the gap.

TAKEAWAY 17W

Electric fields from separate sources add as vectors, so geometry decides whether their contributions reinforce or cancel.

2.3 Inside the sheet

MAST & DOWN-CONDUCTOR · MAST BASE · A ROOM · VERIFY

SCENE 41W

Rigger and mast supervisor Owen Tate, the rigger & mast supervisor, closes the screened-room door. The instrument on the bench inside drops to zero while the flat outside remains near 4 kV/m. Nothing on the instrument rack has been switched off.

GUIDE 51W

Say what the probe inside will read once the door is shut, then shut it and read the probe. The prediction is the part that matters: a number written before the door closes is a claim about how conductors reach equilibrium, and one written afterwards is a description of a display.

ASK 15W

Predict what the bench probe reads with the door shut, then shut it and measure.

BEHIND THE BUTTON 163W

What equilibrium does. Charge in a conductor is free to move, and it moves until the field inside the metal is zero — because any residual field would still be pushing it. The charge ends up arranged on the surfaces, and what a probe inside the enclosure sees is what is left after that rearrangement.

Why it is not perfect. The door seam, the cable entries and the ventilation mesh are apertures, and an aperture lets through a field that scales with its size relative to the wavelengths involved. So the honest prediction is not zero but small — tens of volts per metre against four thousand outside.

Why measuring is the point. The shielding argument is sound and the number it gives depends on hardware nobody derived: how well the door seats, whether the mesh is bonded all round, what the cable glands do. Predicting and then measuring is the only way to find out which of those the enclosure actually has.

TAKES AS READ

charge in a conductor is free to move · gauss's law, and choosing a surface it can be used on — taken as read

VERDICT 149W

Charge in a conductor moves until the field inside the metal is zero, because any field that remained would still be pushing it. That is the whole of electrostatic shielding — not a material property but a consequence of equilibrium, and it is why the probe drops the moment the door seats. What it does not give is exactly zero. The seam, the ventilation mesh and the cable glands are apertures, and each lets a fraction through: measured, the bench reads about 40 V/m against 4 kV/m outside, an attenuation of roughly a hundred. Predicting zero is the right physics and the wrong number, and the gap between them is the hardware. That is why this stop asks for the prediction first and then for the measurement: the argument tells you

> the field inside a closed conductor is zero, and only the probe tells you what your enclosure is.
>
> **TAKEAWAY** 14W
>
> Electrostatic shielding is a consequence of how conductors reach equilibrium in an applied field.

END OF THE DAY 18W

Gauss turns a field into a charge whenever a surface can be chosen the field is simple on.

DAY 3 OF 15

Volts are a path

PLAN CARD 113W

Wednesday. The log has a column for potential difference. It has been empty since the station opened. Vero wants it filled from yesterday's field, not from a textbook. Ingrid Lauwers, the emc engineer, has asked one question. Between which two points. It sounds like fussing. It is not. Volts are always volts between one place and another place. Yusuf Kaya, the records officer, has three shots from last season with no criterion written down. That is how a column stays empty for years. Today you walk the field down a path and get volts out of it. Then you say what a number in that column has to name before it means anything.

BRIEFING 13W

The season log wants the cloud-to-ground potential, and nobody has written the integral.

WHAT YOU DECIDE 13W

Turn a field in volts per metre into the volts between two places.

3.1 Down the path, and the sign

FIELD & CHARGE · FIELD STATION · A ROOM · DERIVE

SCENE 33W

The field under yesterday's cell was 4.5 kV/m, near enough uniform, and the cloud base was 1.2 km up on the radar. Vero wants the potential difference between the base and the flat.

GUIDE 58W

Build it a line at a time. Potential difference is the line integral of the field along a path, with a minus sign in front. The field here is near enough uniform, so the integral collapses to a product. The sign is the part that catches people, so decide which way your path runs before you evaluate anything.

ASK 15W

Get the potential difference between the cloud base and the ground out of the field.

BEHIND THE BUTTON 158W

What the minus sign is doing. Moving against the field costs energy, and potential is energy per unit charge. So a path that runs upward through a downward-pointing field arrives at a higher potential. The sign in the definition is what encodes that, and dropping it gives a cloud base below ground potential.

Why uniform matters. A line integral over a changing field needs the field as a function of position. Here the mills say the field is near enough constant from the flat to cloud base, so the integral is field times distance. That approximation is worth stating, because the real field is not uniform near the mast.

How big the answer is. A kilometre of a few kilovolts per metre gives megavolts. That number is why a rocket trailing a wire can trigger a strike, and why the station's whole safety case is written around the field at the flat rather than around the potential aloft.

TAKES AS READ

the field between the layer and the ground is close to uniform · work per unit charge is what a volt is

VERDICT 75W

The result is −5.4 MV. The sign says the cloud base is at lower potential than the ground under this field direction. Potential is the line integral of the field, while the field is the spatial derivative of potential with a minus sign. Here the field is treated as uniform, so the integral becomes a product. A real cell would require the full integral. Every reported potential must also name the two points being compared.

TAKEAWAY 20W

With a uniform field the integral is a product, and the minus sign is the physics rather than the bookkeeping.

3.2 One foot and the other

COUPLING & THE OUTSTATION · OUTSTATION · A PERSON · BALLPARK

SCENE 39W

EMC engineer Lauwers has the injection-test map on the trailer wall. Rings of equal potential spread from the mast base, crowded near it and widely spaced farther out. She asks where one ordinary stride crosses the largest voltage difference.

GUIDE 64W

Work out the voltage between one foot and the other where the rings are most crowded. The step voltage is the potential gradient there multiplied by the length of a stride. Two tiles are the ground's own potential at those radii, which is a large number and not the one that matters — a bird on a wire sits at kilovolts and is unharmed.

ASK 18W

What voltage appears between one foot and the other at the crowded rings, one metre from the rod?

BEHIND THE BUTTON 156W

What the rings mean. Each is a set of points at the same potential, so walking along one costs nothing and crossing them costs the difference. Current spreading from the mast base falls off with distance, which is why the rings crowd near the rod and open out further away.

Why the gradient is the whole answer. What passes through a person is decided by the difference between their feet, not by the potential of the ground under them. Near the rod that difference over one stride is thousands of volts; out at forty metres the same stride crosses a few tens.

What follows for the site. It is the reason the working rule is to stand still or shuffle during a storm rather than stride, and the reason a buried grading ring is worth its copper: flattening the gradient near the rod is the only way to make the same current survivable to walk near.

TAKES AS READ

current spreading through soil produces a potential that falls with distance · electric potential as a line integral of the field — taken as read

VERDICT 167W

The step voltage is the potential gradient times the stride: V_step = (dV/dr) × d. One metre from the rod the injection test measures a gradient of about 3,800 volts per metre, so an ordinary 0.9 m stride crosses roughly 3,400 V. At forty metres out the gradient has fallen to about 30 V/m and the same stride crosses 27 V. The ground's own potential is a much larger number in both places — 14 kV at one metre — and it is the wrong one to quote, for the same reason a bird on a transmission line is unharmed at hundreds of kilovolts: nothing passes through a body held at one potential. What passes through a person is decided by the difference between contact points. That is why the site rule during a storm is to stand still or shuffle rather than stride, and why a buried grading ring earns its copper: it cannot reduce the current, and it can flatten the gradient the current produces.

TAKEAWAY 22W

What hurts somebody standing on the ground is the difference between where their feet are, not the potential of the ground itself.

3.3 The number that has to exist first

LAUNCH & RECORDS · LAUNCH CONTROL · A ROOM · CHOICE

SCENE 38W

Records lead Yusuf Kaya has three shots from last season with no criterion recorded. The mill logs preserve every launch field, but nobody can reconstruct what threshold the team had committed to before seeing how each afternoon ended.

GUIDE 60W

All four options are claims about when a rule is worth writing. Ask of each whether it depends on knowing how the afternoon ended. Two of them argue that a larger sample is better, which is true of data and not of criteria. A clean shot makes its threshold look reasonable afterwards. So does a cautious hold followed by nothing.

ASK 16W

Why is a criterion written afterwards worth so much less than the same words written before?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a decision can be described after the event by the person who made it

VERDICT 78W

People reconstruct decisions after learning how events ended, often without realizing it. A clean shot makes its launch threshold look more reasonable. A cautious hold followed by nothing makes caution look more reasonable. Neither memory is dishonest, but neither is an unbiased test. Writing the criterion first lets every launch and hold be judged against the same rule. Only then can the season test whether the rule works, instead of quietly changing the rule to fit the outcomes.

TAKEAWAY 15W

A criterion is evidence about a decision only if it exists before the outcome does.

3.4 Twenty-five ohms, at what — Review

EARTHING & TRANSIENTS · EARTHING COMPOUND · A ROOM · CALLBACK · SEQUENCE

SCENE 43W

Blessing Okoro, the test engineer, has the earth tester and two spikes laid out on the tailgate at the compound. She wants the outstation rod measured the same way the April certificate was, and the steps agreed before anybody walks the tape out.

GUIDE 77W

Four steps, and the order decides whether the number belongs to the rod or to the network around it. Ask of each what has to have happened before it. Two of the four are arrangements — what is disconnected, and where the spikes are driven — and two are the measurement and what gets written beside it. An arrangement made late does not slow the test down; it makes the reading describe something other than the rod.

ASK 8W

Order the fall-of-potential test on the outstation rod.

BEHIND THE BUTTON 206W

Why soil is a resistor with no fixed value. The current from an electrode spreads into the ground, and almost all of the voltage appears within the first few metres, where the cross-section it is spreading through is still small. That volume is the electrode's resistance area. What the tester reports is a property of that region and of the path the current happened to take, not a property of the rod's metal.

What the plateau is. Walk the potential probe along the line between the rod and the current spike and the reading climbs, flattens, then climbs again as the spike's own resistance area is entered. The flat part is where neither region is being sampled, and it is the only part of the curve that means anything. Put the spike in too close and the two regions overlap, and no plateau appears at all.

Why the frequency is worth recording. An earth tester works at a low frequency chosen to step over the stray currents already in the soil, which makes its answer a steady-current answer. That is the whole of the argument the April certificate started: the figure is correct, it is honest, and it describes a regime a strike is not in.

TAKES AS READ

current spreading through soil produces a potential that falls with distance · an instrument measures under the conditions it applies, not the ones you care about · current as a flow, and current density — taken as read · electric potential as a line integral of the field — taken as read

VERDICT 84W

Soil is a resistor whose value depends on where the current goes, so the test measures the rod only if the current has nowhere else to travel and nothing it has already raised. Lifting the bond settles the first. Driving the spike well out settles the second, because two overlapping resistance areas give a curve that never flattens.

> The plateau is the rod alone. The frequency belongs on the certificate because the answer is a steady-current answer, and a strike arrives in a microsecond.
>
> **TAKEAWAY** 28W
>
> A resistance measured in soil is a property of the path the current took, so the number is worth exactly as much as the arrangement that produced it.

END OF THE DAY 21W

A potential belongs to a pair of points, and a path is how you get from the field to the volts.

DAY 4 OF 15

Why the tall thing is struck

BEFORE THE DAY — FOLLOW

The cable run, walked once by whoever buried it

Every measurement here comes down a cable, and the trench route is not on the drawing at the accuracy that matters. Stay with them along it. Where they stop is where the cable is shallow, and shallow cable is what gets driven over by the very vehicle bringing the next instrument out.

PLAN CARD 121W

Thursday. Tate reports the mast hissing again. The hiss is corona at the tip: air breaking down in a thin glow around the steel. You can hear it from the base in the quiet before a cell. Vero calls that the station working. Brenner calls it the mast being sixty metres of steel with nothing else near it. Both of them are describing the same piece of arithmetic. Neither has done it. Air stops insulating at some field, and nobody here has written that number down. Today you get the field at the tip out of the field over the flat. Then you decide whether the afternoon's marginal cell is a launch. Vero has argued for a week that it is.

BRIEFING 17W

The mast has been drawing corona since the first cell, and nobody has computed the tip field.

WHAT YOU DECIDE 15W

Get the field at the mast tip, and say what it does to the air.

4.1 Sixty metres up, two centimetres across

MAST & DOWN-CONDUCTOR · MAST BASE · A ROOM · DERIVE

SCENE 34W

The flat is at 4.5 kV/m and the mast is earthed. Its whole length therefore sits at ground potential in air that is not. Tate has the tip radius on the drawing: twenty millimetres.

GUIDE 63W

Build it a line at a time. The mast is earthed, so its whole length sits at ground potential while the air around it does not. Treat the tip as a small sphere: the undisturbed potential at that height tells you the charge, and the same charge gives the field at the tip's own radius. Then say what the air does about it.

ASK 16W

Work out the field at the mast tip, and say what the air does about it.

BEHIND THE BUTTON 165W

Why the tip concentrates the field. For a sphere, potential goes as one over r and field as one over r squared. So the ratio of field to potential goes as one over r, and a small radius at a given potential difference means a large field. Sixty metres of potential difference collapsed onto a twenty-millimetre tip is what makes a mast a lightning terminal.

What the air does. Dry air breaks down at about three megavolts per metre. A tip field above that ionises the air around it, which bleeds charge into the atmosphere as corona and puts the mast's tip current on the station's instruments. The mast is not passive; it is doing something continuously.

Why the sphere is a fair model. A rounded tip on a long conductor is locally spherical, and the field at its surface is set by its own radius rather than by the mast's length. The length enters only through the potential the tip is held away from.

TAKES AS READ

an earthed conductor is all at one potential · the potential of the undisturbed air rises with height above the ground

VERDICT 75W

The tip field is about 1.4×10^7 V/m, against roughly 3×10^6 V/m for dry-air breakdown. The air at the sharp tip therefore begins to ionize. Height sets how much potential the earthed mast reaches into. Tip radius then concentrates that potential difference over a small distance. Double the mast height and the tip field doubles. Make the tip radius ten times smaller and the field rises by ten. Shape is the amplifier.

TAKEAWAY 25W

The field at a conductor's surface is its potential difference from its surroundings divided by its radius of curvature, so sharpness costs more than height.

4.2 What the hiss costs

FIELD & CHARGE · FIELD STATION · A PERSON · CHOICE

SCENE 36W

Halvorsen has two mills to compare: the one at the mast base and the reference one four hundred metres out. Under the same cell the base mill reads consistently lower, and it has done all season.

GUIDE 59W

Four explanations for a mill that reads low, all season, under the same cell. Ask of each whether it would produce a steady difference or an occasional one. Two of them are about earthing and screening, which do not change what the instrument is exposed to. And note which mill people are most tempted to trust: the nearest one.

ASK 17W

Why does the mill at the mast base read lower than the one out on the flat?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

air that has broken down carries charge

VERDICT 79W

Corona sends charge of the tip's sign into the air above the mast. That drifting space charge creates its own field. Near the ground, its contribution partly opposes the cloud field, so the base mill reads low. The effect matters because the nearest instrument is the one people are most tempted to trust. Distance from the mast gives the reference mill cleaner exposure to the ambient field. The difference is therefore a measurement effect created by the charged plume.

TAKEAWAY 20W

A local electric-field sensor can be biased by nearby space charge even when the larger storm field has not changed.

4.3 The afternoon Vero is right about

LAUNCH & RECORDS · LAUNCH CONTROL · A ROOM · CHOICE

SCENE 41W

Brenner checks the launch board twice, then radios that the flat is clear. The incoming cell is smaller than yesterday's and the mills are climbing steadily. Vero looks from the written rule to the rail and asks for the launch call.

GUIDE 56W

All four options describe something true of this afternoon. Ask of each whether it appears in the rule that was written before the season. Two of them compare today with yesterday, which the rule does not do. A criterion nobody uses is not an operating rule, and this is the first cell where it says go.

ASK 10W

What makes this afternoon a launch rather than a hold?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

the launch rule sets thresholds for actions with lead times

VERDICT 84W

The rule exists so this decision is not reinvented under a storm. It was written before the season and states what must be true. Today every stage is satisfied in order, with its required lead time intact. Vero is not claiming that marginal cells are harmless. Her point is narrower: a criterion that nobody uses is not an operating rule. The shot goes. It returns the cleanest waveform of the season so far, and Brenner still has the same rule for the next cell.

TAKEAWAY 20W

A precommitted operating rule should be applied consistently on marginal cases unless new evidence shows the rule itself is inadequate.

4.4 Two plates, and a factor of two — Review

IMPULSE HALL · A ROOM · CALLBACK · BALLPARK

SCENE 43W

Josefa Ruiz, the impulse hall technician, has one plate of the pair standing on its own trestle while the other is away for a new insulator. Its charge per square metre is known from the last shot. Strand wants the field beside it.

GUIDE 69W

One plate, not two. The board carries the plate's charge per square metre, the permittivity of free space, and three quantities that belong to other stops. Pick the two a charged sheet needs, then decide which of the two sheet results this is. The pair in the hall has a second plate driving field the same way across the gap; this plate has nothing across from it at all.

ASK 8W

What field stands beside the single charged plate?

BEHIND THE BUTTON 191W

Why the distance is not in the answer. A pillbox through a sheet has flux out of its two faces and none through its sides, and neither the height of the box nor its area survives the algebra. So the field of a wide sheet is the same a centimetre away and a metre away, which is what makes a plate hazardous to stand near and a cloud base readable from the ground.

What the second plate does. Bring back the other plate with the opposite charge and its field points the same way as the first one everywhere between them, so the two add. That is superposition rather than a new law, and it is also why the field outside the pair nearly vanishes: out there the two contributions point opposite ways.

Why the hall cares which case it is in. The insulation is specified against a field, and a factor of two is the difference between a margin and a flashover. Ruiz measures the standing plate on its own precisely because the answer for a lone sheet is the one that applies while the other is out for repair.

TAKES AS READ

fields from separate charges add as vectors · a plate far wider than the distance of interest can be treated as unbounded

VERDICT 97W

A charged sheet drives field out of both of its faces, so the pillbox has flux through two of them and the result is σ divided by twice the permittivity of free space — about 2,300 volts per metre here. Bring the second plate back carrying the opposite charge and its field points the same way across the gap, so the two add and the answer doubles. Neither result contains the distance from the plate or the area of it. That is why a lone sheet and a pair are two different questions with the same σ.

TAKEAWAY 22W

Which sheet result applies is settled by how many sheets are contributing, not by how much charge any one of them carries.

END OF THE DAY 22W

A conductor takes the potential of the air it reaches into, and a small radius turns that potential into an enormous field.

The cloud as a capacitor

PLAN CARD 117W

Friday. Strand has taken over the board in the hall. The station keeps describing the cell overhead as a charged plate above a flat that conducts. Strand wants the capacitance of that pair written down before Monday. Capacitance is how much charge the pair holds for each volt across it. On Monday somebody will ask for the stored energy, and the energy cannot be had without it. Sifuentes cares more about the trench. Two conductors run side by side down it for two hundred metres. That is the same model with different numbers in it. Today you derive a capacitance from a geometry. Then you find out what filling the gap with an insulator does to it.

BRIEFING 14W

Strand wants the cell modelled as a plate pair before anybody talks about energy.

WHAT YOU DECIDE 13W

Get a capacitance out of the geometry, and see what it is worth.

5.1 Ten square kilometres, twelve hundred metres up

IMPULSE HALL · A ROOM · DERIVE

SCENE 33W

Strand puts the cell on the board as one plate. Ten square kilometres of charged base sit twelve hundred metres above a flat that conducts well enough to serve as the other plate.

GUIDE 53W

Build it a line at a time. Capacitance is charge over potential difference, so the route runs through the field: put charge on the plate, get the field between them, integrate the field to get the voltage, then divide. Every step follows from the one above, and each wrong branch is legal algebra.

ASK 10W

Get the capacitance of the cloud-ground pair from its geometry.

BEHIND THE BUTTON 161W

Why a cloud and the ground make a capacitor. Two conductors with charge on one and an equal opposite charge on the other, separated by an insulator, is what a capacitor is. The flat conducts well enough to serve as the second plate. Whether the geometry is a parallel plate is then a question about area against separation.

Why ten square kilometres over 1.2 kilometres is close enough. Fringing at the edges matters when the separation is comparable with the plate size. Here the plate is about three kilometres across and the gap is 1.2 kilometres, so the parallel-plate result is an estimate with real error rather than an exact answer. Saying which it is matters more than the digits.

What the number is for. Capacitance times voltage is the charge available, and half of C times V squared is the energy. Those set what a strike can deliver, which is what everything downstream of this stop is trying to survive.

TAKES AS READ

the field between oppositely charged plates is σ over ε_0 · capacitance is the charge held per volt between the conductors

VERDICT 103W

The cloud-ground pair comes out near 74 nF from geometry and ε_0 alone. That is what capacitance means: the geometry fixes the ratio of stored charge to voltage before the system is charged. The derivation starts with Gauss's law for the plate field. The path integral then gives the voltage across the separation. Only after those steps does the ratio Q/V become C. Keeping the factor of two straight between one sheet and two plates is essential. The model is a parallel plate capacitor: area of the plates over plate separation, with the cloud as a plate and the flat as the other.

TAKEAWAY 22W

Capacitance depends only on the shape and spacing of the conductors, which is why it can be computed before anything is charged.

5.2 Two conductors, two hundred metres

EARTHING & TRANSIENTS · EARTHING COMPOUND · A PERSON · BALLPARK

SCENE 41W

The armoured signal cable and bare copper earth conductor run side by side for two hundred metres, forty centimetres apart. Neither is connected to the other. Lauwers wants the pair treated as two conductors before anyone calls the trench electrically quiet.

GUIDE 61W

Treat the pair as what it is and put a number on it. Capacitance is charge divided by the voltage that charge produces, so pick the charge the injection test moved onto the earth conductor and the voltage it reached. One tile is the length of the run, which is what the answer scales with rather than part of the ratio.

ASK 11W

What is the capacitance between the two conductors in the trench?

BEHIND THE BUTTON 140W

Why a ratio and not a formula. Capacitance is defined as charge over potential difference, whatever the shape of the conductors. The geometry formulas for parallel plates and coaxial cables are consequences of that definition for particular shapes, and a trench is neither of those shapes.

What it means for the trench. Two conductors with a capacitance between them exchange current whenever the voltage between them changes, and a strike changes it in microseconds. No metallic connection is required, and none of the continuity tests anybody runs would find the path.

Why the length matters. Capacitance per metre is set by the spacing and the material between; the total is that figure times the run. Two hundred metres of trench is a couplant nobody drew on the schematic, and shortening the parallel run is the only intervention that reduces it.

TAKES AS READ

any two conductors separated by an insulator have a capacitance between them

VERDICT 156W

Capacitance is $C = Q / V$, a ratio before it is a formula. The injection test pushed 1.2 microcoulombs onto the earth conductor and the pair reached 500 V, so $C = 1.2 \times 10^{-6} \div 500 = 2.4$ nF over the whole 200 m run — about 12 picofarads a metre, which is what the spacing and the soil between them give. Two things follow. Nothing is connected, so no continuity test finds this path, and it is the path anyway: the current through a capacitance is $C\,dV/dt$, and a strike changes the voltage between these conductors by tens of kilovolts in a couple of microseconds. Put those numbers together and the trench passes tens of amps into a signal cable that nobody wired to anything. And the length is the lever — the capacitance is per metre, so the only intervention that reduces it is running less of the pair in parallel.

TAKEAWAY 11W

> Conductors can exchange transient electrical influence without a direct metallic connection.

5.3 Filling the gap

COUPLING & THE OUTSTATION · OUTSTATION · A ROOM · CHOICE

SCENE 30W

Lauwers has two identical cable samples on the bench. One is insulated with air and one with polyethylene, and the polyethylene one measures more than twice the capacitance per metre.

GUIDE 63W

Four explanations for more than double the capacitance. Ask of each whether it changes the charge, the geometry, or the voltage. The two samples are identical apart from what fills the gap. So nothing about spacing or area has moved. What the material does is shift its own bound charge against the applied field, and that lowers the volts for the same charge.

ASK 10W

Why does filling the gap with polyethylene raise the capacitance?

BEHIND THE BUTTON 195W

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TAKES AS READ

an insulator contains charge that can shift slightly without leaving its molecules

VERDICT 75W

A dielectric polarizes in an electric field. Bound positive and negative charge shift slightly in opposite directions, creating a field that opposes the applied field. For the same free charge, the net field and voltage fall. Because $C = Q/V$, capacitance rises by the material's dielectric factor, about 2.3 for polyethylene here. Polyethylene also has a larger dielectric strength than air. That separate property lets the cable survive a larger field before electrical breakdown begins.

TAKEAWAY 12W

> The material between conductors changes the relationship between stored charge and voltage.

5.4 A spinning shutter and a sign — Review

FIELD & CHARGE · FIELD STATION · A ROOM · CALLBACK · PROTOCOL

SCENE 41W

Field-instruments technician Mira Halvorsen has the reference set for the mill out on the bench: two steel spheres of the same size, one solid and one hollow, and a small charged ball on a silk thread. Nothing has been charged yet.

GUIDE 75W

Four arrangements and four descriptions of where the charge ends up. Ask of each arrangement which surfaces exist: a hollow sphere has an inner one as well as an outer one, and a solid sphere has only the outside. Then ask whether the ball has touched anything, because that decides whether what it carried is still its own. One description goes to each arrangement, so settling the two you are sure of constrains the rest.

ASK 9W

Match each arrangement to where its charge ends up.

BEHIND THE BUTTON 212W

Why a conductor cannot hold charge in its interior. Charge in a metal is free to move, and it goes on moving for as long as any field is pushing it. Equilibrium is the state where nothing is pushing, which means the field inside the metal is zero. Gauss's law on a surface drawn just inside the metal then says the charge enclosed there is zero, so whatever the conductor carries has to be on a boundary.

What a cavity adds. A closed cavity gives the conductor a second surface, and the two behave differently. Charge lowered into the cavity without touching anything draws an equal and opposite amount onto the cavity wall and pushes the same amount out onto the outside. Seen from beyond the sphere the arrangement is invisible: the outer surface reports the total and says nothing about how it got there.

Why this is the mill's problem too. The plate the shutter uncovers is a conductor standing in somebody else's field, and what the electronics measure is the charge induced on that plate's own face. Knowing that charge lives on surfaces, and that an enclosure has an inner surface of its own, is what lets a mill be checked against a known field rather than against another mill.

TAKES AS READ

charge in a conductor is free to move · flux out of a closed surface counts the charge inside it

VERDICT 100W

Charge in a metal is free to move, and it moves for as long as a field is pushing it. Equilibrium is the state where nothing is pushing, so the field inside the metal is zero. Draw a Gaussian surface just inside the metal and the charge it encloses must be zero too, which puts everything the sphere carries on a boundary. A solid sphere has one boundary. A hollow sphere has two, and an empty cavity stays field-free. A

> charged ball lowered inside draws an equal and opposite charge onto the cavity wall and pushes its own share out.
>
> **TAKEAWAY** 21W
>
> Charge on a conductor ends up on a surface, because anywhere else there would still be a field left pushing it.

END OF THE DAY 19W

Capacitance is a ratio before it is a formula, and the formula is Gauss followed by a line integral.

Where the half comes from

PLAN CARD 138W

Monday of week four. Josefa Ruiz, the impulse hall technician, wants the hall's energy figure checked. The season's biggest impulse test runs on Friday. The hall holds twelve stages. Each stage is a hundred nanofarads, charged to fifty kilovolts. The placard on the barrier says fifteen hundred joules. It has said that since the hall was built, by people who have all since left. Strand's view of it is blunt. A number nobody has re-derived is a rumour with a unit after it. He has been saying that about this one for two seasons. Today you get the stored energy out of the work it took to put the charge there. Ruiz wants the earthing stick explained in the same breath. You also decide what the hall can honestly stand in for when the sky will not cooperate.

BRIEFING 15W

Twelve stages at fifty kilovolts, and the safety case rests on a number nobody re-derives.

WHAT YOU DECIDE 14W

Derive the energy in the bank, and say what it means for the hall.

6.1 Fifteen hundred joules, or three thousand

IMPULSE HALL · A ROOM · DERIVE

SCENE 33W

Impulse-hall technician Ruiz sets one stage on the bench: 100 nF charged to 50 kV. The bank placard says 1,500 J for twelve stages, and Strand wants the number derived rather than trusted.

GUIDE 56W

Build it a line at a time. Energy stored is the work done charging the capacitor, and the work of moving each increment of charge is V dq. The voltage rises as the charge accumulates, so this is an integral rather than a product. Finish with joules for one stage, so the placard can be checked.

ASK 11W

Work out the energy stored in one stage of the bank.

BEHIND THE BUTTON 147W

Why not simply QV. Pushing the last coulomb on costs the full final voltage; the first one costs almost nothing. Integrating V dq with V = q/C gives half of QV, and the missing half is the reason a placard that quotes QV is out by a factor of two.

Where the placard number comes from. Twelve stages at the derived energy each gives the total. If that does not match the printed 1,500 joules, either the placard is quoting a different charging voltage or somebody used the wrong formula. Deriving it is how you find out which.

Why it matters on the bench. The energy in one stage is what a person standing beside it would receive if it discharged through them. A factor of two on that number is the difference between two safety categories, which is why Strand wants it derived rather than trusted.

TAKES AS READ

moving charge onto a capacitor takes work against the voltage already across it · the voltage across a capacitor is proportional to the charge on it

VERDICT 78W

One 100 nF stage at 50 kV stores 125 J, so twelve stages store 1,500 J. The factor of one-half is not a convention. As charge is added, the capacitor voltage rises from zero to its final value. Early charge crosses little voltage; the last increment crosses almost all of it. Integrating V dq therefore uses an average voltage of one-half the final value. Dropping that factor doubles the energy and turns a derivation into a dangerous overestimate.

TAKEAWAY 21W

Charging is done against a voltage that grows as you go, so the work is an integral and not a product.

6.2 Charge that is still there

MAST & DOWN-CONDUCTOR · MAST BASE · A PERSON · ATTEST

SCENE 35W

Ruiz keeps the barrier closed even after the supply is off and the meter reads zero. The earthing stick must be connected to the bank and visibly left in place before anyone enters the hall.

GUIDE 57W

Five signed statements about the state of the hall, and two verifications. Each one is somebody's record rather than a condition you have seen, so decide which would hurt most if the paper were wrong and check those. A claim can only be held once it has been verified, and holding everything critical is not caution here.

ASK 17W

Which of these signed claims about the hall cannot be held, given what the evidence actually supports?

BEHIND THE BUTTON 146W

Why a meter reading zero is not a discharged bank. A panel meter reads what its own circuit sees, and a bank whose bleed path has failed can sit at full charge behind an open contactor with nothing on the front panel to say so. The stored energy is $\frac{1}{2}CV^2$, and it does not decay because the supply was switched off.

What the stick actually provides. A visible, mechanically held connection between the bank terminal and earth — the discharge path itself, not a statement about one. Its whole point is that it can be seen from the door by somebody who read no paperwork.

Why the switch position is the weakest kind of evidence. An isolator handle records what somebody moved, and a contactor that has welded shut moves nothing while showing the same handle position. The condition and the record come apart exactly there.

TAKES AS READ

a capacitor holds its charge once the supply is removed

VERDICT 164W

The hall is safe when there is a discharge path somebody can see, not when the paperwork says the supply is off. Two of these claims are critical and unbacked. The bleed resistor is signed off on a 2019 commissioning sheet, and a bleed path that has failed open leaves the bank at full charge with a panel meter reading zero — the meter's circuit and the bank's terminals are not the same node. The isolator claim records a handle position, and a welded contactor holds the same handle while passing current. Two others are critical and genuinely backed: the earthing stick is visible from the door, which is the point of it, and the terminal voltage was measured directly this morning at the bank rather than inferred from the panel. The stored energy is $\frac{1}{2}CV^2$ and nothing about switching the supply off removes it. That is why the safe state is a connection you can see and not a signature you can read.

TAKEAWAY 19W

Stored electrical energy can remain after its source is removed, so a safe state requires a controlled discharge path.

6.3 A clean pulse and a real one

LAUNCH & RECORDS · LAUNCH CONTROL · A ROOM · CHOICE

SCENE 40W

Strand can fire the bank into the mast on any afternoon, with a rise time he chooses. Vero would rather wait for a cell. Kaya has to write down which of the two the season's waveform library is built from.

GUIDE 63W

Four things the hall might offer that a natural stroke cannot. Ask of each whether the hall's advantage is control or magnitude. The hall chooses the moment and the shape, every time. A natural stroke varies from event to event, and that variation is what the station exists to measure. So the library needs both kinds labelled, or calibration data pass as weather.

ASK 14W

What can the hall's impulse do for the station that a triggered strike cannot?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

an instrument is calibrated against a signal whose shape is known

VERDICT 82W

The impulse hall gives a signal whose timing and waveform are chosen in advance. That makes repeated calibration possible. A natural triggered stroke is different: its peak, rise time and number of strokes vary from event to event. Those variations are exactly what the station exists to measure. The hall can therefore test response and timing under repeatable conditions. It cannot replace the weather record. The library needs both kinds of shot labelled clearly so calibration data never masquerade as atmospheric data.

TAKEAWAY 12W

> Calibration data and natural-event data answer different questions: repeatability versus real-world variation.

6.4 One foot and the other — Review

COUPLING & THE OUTSTATION · OUTSTATION · A ROOM · CALLBACK · BALLPARK

SCENE 42W

Sam Abioye, the instrumentation technician, has the injection set on the outstation's own earth rod, the one with no grading ring around it. The rings on his plot crowd hard at two metres. Lauwers wants the number a stride would cross there.

GUIDE 61W

The board carries the gradient two metres out, the length of a stride, and three numbers that belong to the site rather than to anybody's feet. One of them is the potential the ground itself sits at, which is the largest figure on the board and the wrong one. Pick the pair whose product is a voltage across a single stride.

ASK 12W

What voltage does one stride cross two metres from the ungraded rod?

BEHIND THE BUTTON 194W

What the rings are. Each is a set of points at the same potential, so a foot moved along one crosses no voltage at all and a foot moved across them crosses the difference. The rings crowd where the current is spreading through the least ground, which is close to the rod, and they open out as the cross-section grows.

Why an ungraded rod is the worse case. A grading ring cannot reduce the current the rod has to pass, and it does not try to. What it does is spread the same potential drop over more distance, which flattens the gradient near the rod. The outstation rod has none, so the whole drop happens in the first few metres, which is exactly where anybody working on the trailer stands.

Why the ground's own potential is the wrong number to quote. A body at one potential has nothing driving current through it, which is the whole reason a bird on a transmission line is unharmed at hundreds of kilovolts. What decides the current through a person is the difference between their contact points, and on a flat that means the distance between their feet.

TAKES AS READ

current spreading through soil produces a potential that falls with distance · a potential difference across two contact points is what drives current through a body · electric potential as a line integral of the field — taken as read

VERDICT 87W

The rings are contours of equal potential, so walking along one crosses nothing and crossing them costs the difference. Two metres from an ungraded rod the injection test gives a gradient near 1,600 volts per metre, and an ordinary stride of nine tenths of a metre crosses about 1,400 volts. Twenty metres out the gradient has fallen to twenty volts per metre and the same stride crosses eighteen. The ground's own potential is larger in both places, and nothing passes through a body held at one potential.

TAKEAWAY 29W

> A grading ring cannot make the current smaller, and flattening the gradient it produces is the only thing that makes the ground beside a rod safe to walk on.

END OF THE DAY 23W

The energy in a capacitor is the work done against the voltage that was already there, which is where the half comes from.

DAY 7 OF 15

Three instruments that agree

PLAN CARD 112W

Tuesday. Kenji Nakata, the current measurement engineer, has the S-114 record open. The funding review is going to quote it. Three separate measurements of last August's strike agree to within ten per cent. The station has treated that as confirmation for a year. Vero has already put it in a draft. Nakata does not like how tidy it is. Ana Sifuentes has noticed something and has not made much of it yet. Every digitiser on the site is referred to one earth rod. Today you open each channel and find what its reading was measured against. Then you work out what the agreement between them is worth to a reviewer who asks.

BRIEFING 15W

Three shunts, three readings within ten per cent, and one thing they have in common.

WHAT YOU DECIDE 11W

Find out whether the current measurements are independent of each other.

7.1 Agreement is not independence

MAST & DOWN-CONDUCTOR · MAST BASE · A ROOM · TRACE

SCENE 36W

Five records exist of strike S-114. Four of them give a peak current and the fifth gives a stroke count. Nakata has their derivations closed and wants the shared piece named before anybody quotes the average.

GUIDE 53W

Open a channel and the panel shows what its reading was referred to. Tick the ones that would still stand if a shared reference were wrong, and untick the rest. Then name the piece they have in common. Four records agreeing is not four independent measurements, and both halves of this are graded.

ASK 24W

Four of these five report the same peak. Open them, keep whatever still stands on its own, and name what they have in common.

BEHIND THE BUTTON 143W

Why agreement is not independence. Four peak currents computed through the same rod reference will agree with each other whatever that reference does, because the error is common to all of them. Averaging them narrows the scatter and does nothing about the bias, which is the most convincing way to be wrong.

What the fifth record is doing. A stroke count is not a current, so it cannot be averaged with the others. It is on the board because it does not pass through the suspect step, which makes it worth more than any of the four for this particular question.

Why naming the shared piece is the answer. Once the common reference is identified it can be corrected, and every dependent record moves together. Withdrawing the average instead leaves the same error in place and tells the review that nothing is known.

TAKES AS READ

a measurement is made against something, and that something can be wrong · current as a flow, and current density — taken as read

VERDICT 80W

A shunt reports voltage across a known resistance, and voltage is always a difference between two points. If several channels share the same earth rod, motion of that reference appears in all of them together. The channels can then agree without being independent. The clamp is different. It responds to the magnetic field around the conductor and never uses the earth rod. That separate dependency chain is what makes its agreement informative rather than another copy of the same reference.

TAKEAWAY 16W

A number quoted three times from one reference is one number, however many instruments produced it.

7.2 The rod is not a fixed point

EARTHING & TRANSIENTS · EARTHING COMPOUND · A ROOM · CHOICE

SCENE 32W

Sifuentes draws the rod as a resistor rather than as a symbol. Thirty kiloamps into twenty-five ohms puts the whole compound somewhere, and every instrument in the rack calls that somewhere zero.

GUIDE 63W

Four readings of what the rod does during a strike. Ask of each whether the earth is being treated as a conductor with resistance, or as a fixed zero. Push current through a resistance and a potential difference appears, for as long as it lasts. Everything on the site calls that potential zero. Whether that matters depends on how many places are earthed.

ASK 15W

What happens to the potential of the earth rod while the strike current is flowing?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

a resistance carrying current has a voltage across it · current as a flow, and current density — taken as read

VERDICT 87W

The earth is a conductor with resistance, so pushing current into it produces a potential difference exactly as a resistor does. Thirty kiloamps through twenty-five ohms lifts the rod by hundreds of kilovolts relative to soil far away, for as long as the current lasts. That is not necessarily damaging: if everything on the site is bonded to the same rod, everything rides up together and nothing sees a difference. The damage starts wherever two things are earthed at two places, which is what the outstation is.

TAKEAWAY 19W

> An electrical reference is only a chosen zero; real resistance can make that reference move when large current flows.

7.3 One peak, one shared offset

LAUNCH & RECORDS · LAUNCH CONTROL · A PERSON · DEGENERACY

SCENE 35W

Yusuf Kaya has three rod-referenced shunt peaks at 28, 29 and 31 kA, plus a 30 kA clamp reading. The review wants one peak, but the three shunts still share the same moving earth reference.

GUIDE 58W

Two controls slide along a family of solutions that fit the shunt mean equally well. Find out how wide that family is before deciding anything. Then bring in the clamp reading, which does not share the shunts' earth reference, and watch what it does to the family. Report the peak when the data supports one rather than several.

ASK 14W

How many true peak currents fit the shunts before the independent clamp is added?

BEHIND THE BUTTON 152W

Why three shunts do not settle it. All three are referenced to the same moving earth, so a shift in that reference moves all three together. The mean of 28, 29 and 31 is therefore consistent with a range of true peaks paired with a range of offsets — one equation, two unknowns.

What the clamp brings. It measures the current a different way, without the rod reference. So it constrains the same true peak through a different combination of unknowns, and where the two constraints cross is the answer. That is why one independent measurement is worth more than three more shunts.

Why saying 'several' is a legitimate answer for a while. Reporting a single peak from the shunts alone would be reporting a number the data does not contain. The honest intermediate answer is the family, and the professional move is to go and get the measurement that collapses it.

TAKES AS READ

an average of several readings is only better than one if they are independent

VERDICT 84W

The three shunts do not independently determine the true current because each is measured against the same earth rod. A shift in that rod can move all three together. The shunt average therefore permits several combinations of true peak and shared reference offset. The magnetic clamp uses different physics and no earth reference. Its 30 kA reading cuts across that family of solutions. The important gain is not another decimal place. It is removing an ambiguity that three agreeing channels cannot remove by themselves.

TAKEAWAY 17W

A shared reference can trade off against the true signal; an independent measurement can collapse that ambiguity.

END OF THE DAY 13W

Instruments that share a reference share its errors, and their agreement measures nothing.

DAY 8 OF 15

What thirty kiloamps does at two metres

BEFORE THE DAY — HUNT

Six earth bonds, and the sheet accounts for four

Every instrument on the mast is bonded, and two bonds are not where the sheet says. A bond nobody can find is either doing nothing or carrying strike current through a signal cable, and both possibilities end the season for that channel.

PLAN CARD 111W

Wednesday. Tate has been asked to move the instrument cabinet. He wants a reason he can put on a job sheet. The cabinet has stood two metres from the down-conductor since the mast went up. Nakata's clamp says thirty kiloamps went past it last August. Nobody has ever written down what that means at the cabinet door. A current makes a magnetic field around itself. That field pushes on any other current near it. Today you get the field out of Ampère's law, and you put a number on the force it lands on the bonding straps. Then you settle what the hall can test about this, and what it cannot.

BRIEFING 15W

The instrument cabinet sits two metres from the down-conductor and has done for nine years.

WHAT YOU DECIDE 14W

Get the field around the down-conductor, and say what has to live in it.

8.1 Once round the conductor

MAST & DOWN-CONDUCTOR · MAST BASE · A ROOM · DERIVE

SCENE 30W

Thirty kiloamps down a single vertical conductor, and a steel cabinet two metres from it. Tate wants the field at the cabinet in a unit he can compare with something.

GUIDE 53W

Build it a line at a time. Ampère's law relates the field around a closed loop to the current through it. Choose a circle centred on the conductor and the symmetry makes the integral trivial. Evaluate at two metres and give the answer in tesla, so it can be compared with something familiar.

ASK 16W

Get the magnetic field two metres from the down-conductor while the strike is at its peak.

BEHIND THE BUTTON 151W

Why a circle. The field around a long straight conductor circles it, with the same magnitude everywhere at a given radius. So the line integral around a concentric circle is the field times the circumference, and Ampère's law hands you the field in one step. Any other loop gives the same physics and much worse algebra.

What to compare it with. The Earth's field is about fifty microtesla. A steel cabinet two metres from thirty kiloamps sits in something far larger for a few microseconds, which is why the cabinet's contents care and why the comparison is the useful part of the answer.

Why the peak matters rather than the average. The field follows the current instant by instant, so the number worth quoting is at the peak. It is also the moment the induced voltages downstream are largest, which is the subject of the trench-loop stop later in the fortnight.

TAKES AS READ

the circulation of the magnetic field around a closed path counts the current through it · the field of a long straight conductor circles it · current as a flow, and current density — taken as read

VERDICT 77W

At two metres, a 30 kA current produces about 3.0 mT. Ampère's law works cleanly because a circular path around the conductor has B parallel to the path and nearly constant in size. That symmetry lets B leave the integral. The result falls as $1/r$, not $1/r^2$, because the source is treated as a long line current. The magnetic field also stores energy while the current flows, which is why inductance later matters to fast strike transients.

TAKEAWAY 19W

Choosing a path the field is constant along is what turns Ampère's law from a statement into a number.

8.2 How much charge came down

IMPULSE HALL · A PERSON · BALLPARK

SCENE 38W

Strand wants the August stroke written in coulombs as well as kiloamps. The clamp recorded a 30 kA peak and the current stayed near that scale for about 100 microseconds. The hall comparison is waiting on the number.

GUIDE 58W

Five numbers, and three of them belong to other parts of the event. The rise time of the front. The earth resistance. The energy in the impulse store. Ask of each whether charge depends on it. Peak current, total charge and rise time describe different things. The front controls induced voltage. The duration controls how much charge moves.

ASK 9W

Estimate the charge delivered by the August return stroke.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

charge is current multiplied by the time it flows for

VERDICT 81W

The stroke moves about 3 C in roughly 100 microseconds. That is much more charge than the impulse hall bank can deliver, even when the hall reproduces a similar front rate. Peak current, total charge and rise time describe different parts of the event. The microsecond front controls induced voltage through dI/dt. The longer current duration controls how much charge moves. A useful calibration must therefore say which property it reproduces rather than calling the entire laboratory pulse equivalent to lightning.

TAKEAWAY 22W

Peak current, delivered charge and rise time describe different parts of a pulse, so matching one does not match the whole event.

8.3 The cabinet is a loop

COUPLING & THE OUTSTATION · OUTSTATION · A ROOM · BALLPARK

SCENE 34W

Lauwers is unbothered by three millitesla as a number and very bothered by how fast it arrives. Inside the cabinet there are half a dozen circuits, each of them a loop of some area.

GUIDE 61W

Work out the field the down-conductor puts at the cabinet, from the current in it and the distance to it. A long straight conductor gives a field that falls as one over the distance, so pick the current, the distance and the constant. One tile is the cabinet's own loop area, which belongs to the next step rather than this one.

ASK 13W

What magnetic field does the down-conductor put at the cabinet during the stroke?

BEHIND THE BUTTON 161W

Where the field comes from. A long straight conductor carrying current produces a circulating field of magnitude $\mu_0 I / 2\pi r$ — the form the Biot–Savart integral takes for a straight wire, and the one worth remembering because most conductors on a mast are straight over the distances that matter.

Why the rate matters more than the number. Three millitesla is unremarkable. Arriving in two microseconds is not: the voltage driven around a loop is the area times the rate of change of the field through it, so the same field arriving a thousand times faster drives a thousand times the volts.

What that does inside the cabinet. Every circuit in there encloses some area, and none of them was drawn as an antenna. A loop of a tenth of a square metre in a field changing at fifteen hundred tesla a second sees a hundred and fifty volts around it, which is what nobody wired and everybody has to live with.

TAKES AS READ

a changing magnetic field through a loop drives a voltage around it

VERDICT 220W

For a long straight conductor the field is $B = \mu_0 I / 2\pi r$. With a 30 kA stroke down a conductor 2 m from the cabinet, $B = (4\pi \times 10^{-7} \times 30{,}000) \div (2\pi \times 2) \approx 3.0 \times 10^{-3}$ T, which is the three millitesla on the sheet. That number on its own is harmless. What matters is that the stroke delivers it in about two microseconds, so dB/dt is roughly 1,500 tesla per second, and the voltage driven around a loop is the flux rate through it: $\text{EMF} = A\, dB/dt$. Every circuit in the cabinet encloses some area, and a tenth of a square metre gives about 150 V around a loop nobody drew as an antenna. No connection to the down-conductor is needed for any of this. That is why the fix is loop area rather than insulation — twisted pairs, tight returns, and cable routes that do not enclose the cabinet's own volume. The same field acts on the conductors carrying it: the force on a current in a field is IlB, so two parallel bus bars in a stroke repel hard enough to need a brace. And the energy stored in the inductance — half L

I squared, tens of kilojoules in the strike — has to go somewhere when the current stops.

TAKEAWAY 16W

A changing electromagnetic field can drive voltage in a nearby circuit without any direct electrical contact.

8.4 The charge, while the hall waits

IMPULSE HALL · A ROOM · HOLD

SCENE 41W

The bank has to sit at its charging voltage while the crew finish in the hall and the trigger is armed. The bleed resistors are draining it the whole time, and the charging set is the only thing putting it back.

GUIDE 51W

Hold the bank voltage inside the band on the console. The band narrows as the shot approaches, because the peak current depends on where the bank sits when the trigger fires, and a shot fired off-voltage is a shot nobody can compare with the others. The charging set is your control.

ASK 11W

Hold the bank at its charging voltage until the shot fires.

BEHIND THE BUTTON 121W

Why a bank drains at all. The bleed resistors are a safety feature, not a fault: they are what makes the bank safe to approach after a shot, and they work all the time, including now.

Why that is a rate. The bank loses charge continuously through the bleed path, so the voltage falls and keeps falling. A charging set switched on and off leaves the bank somewhere it did not intend to be.

Why the margin tightens near the shot. The energy stored is proportional to the square of the voltage, so a few per cent of voltage is a larger error in energy — and the whole point of the shot is comparison with the others in the series.

TAKES AS READ

a charged bank stores energy and can lose it through a resistor

VERDICT 155W

The bleed resistors are draining the bank the entire time it is charged, which is exactly what they are for — they are the reason anybody can walk into the hall after a shot. But they do not wait for the shot, so the bank is losing charge continuously while the crew finish and the trigger is armed. That makes the charging set a rate to be matched rather than a switch to be flicked: set it to replace what the bleed takes, and the voltage sits still. The band narrows because the stored energy goes as the square of the voltage, so a small sag near the trigger is a larger error in the energy delivered, and the value of this shot is that it can be compared with the others in the series. A shot fired two per cent low is not a slightly smaller shot; it is a point nobody can plot.

TAKEAWAY 17W

A store that is always leaking is held by matching the leak, not by topping it up.

END OF THE DAY 15W

A current does not stay in its conductor as far as anything nearby is concerned.

Nothing was connected to anything

PLAN CARD 103W

Thursday. This is the question the station has argued about for a year. Last August every card in the outstation trailer died. The trailer sits two hundred metres from the mast. It is earthed to its own rod. No conductor runs between it and anything the strike touched. Sifuentes has defended that earthing for twelve months. Lauwers keeps saying the earthing was never the route in. They cannot both be right, and the review will ask. Today you compute the flux through the cable loop that runs beside the down-conductor. Then you find out what one microsecond of rise time does to it.

BRIEFING 16W

The outstation cards died in August, and the trailer is bonded to nothing that was struck.

WHAT YOU DECIDE 11W

Work out how a strike reached a cable it never touched.

9.1 The loop nobody drew

COUPLING & THE OUTSTATION · OUTSTATION · A ROOM · DERIVE

SCENE 37W

Sam Abioye, the instrumentation technician, has the outstation trench route on the wall. For forty metres, the signal pair runs beside the down-conductor. Its near side is 1.5 m away; the far side is 3.5 m away.

GUIDE 57W

Build it a line at a time. The field from the down-conductor falls as one over the distance, so the flux through the trench loop needs an integral across it, from the near side to the far side. Then Faraday's law turns the changing flux into a voltage. Evaluate it at the front of the August stroke.

ASK 13W

Work out the voltage driven around the trench loop by the August stroke.

BEHIND THE BUTTON 164W

Why an integral rather than a product. The field is not uniform across the loop: at 1.5 metres it is more than twice what it is at 3.5. Flux is the field integrated over the area, and with B going as one over r the integral gives a logarithm of the ratio of the two distances.

Why the ratio is what matters. Only b over a appears, so moving both sides of the pair further out changes the answer far less than separating them does. A signal pair run tight together in the same trench sees very little; the same pair split by two metres sees a great deal.

What the voltage does. It appears in series with the signal, so the instrument at the far end cannot distinguish it from the measurement. That is the mechanism behind most of the station's unexplained records, and it is why the trench route on the wall is a wiring diagram and a physics problem at once.

TAKES AS READ

the field around a long conductor falls off as one over the distance · a changing flux through a loop drives an EMF around it

VERDICT 80W

The trench loop sees about 200 kV even though the strike never touches it. Integrating the magnetic field across the loop gives a mutual inductance of 6.8 μH. Multiplying that by the strike's dI/dt produces the induced EMF. The logarithm comes from the field being stronger on the near side than the far side. Lauwers circles the answer on the board. The failed path was geometric coupling through a loop, not current arriving through the cable insulation or trailer earth.

TAKEAWAY 25W

The field across the loop is not uniform, so the flux is an integral, and it comes out as a logarithm of the two distances.

9.2 What survived in August

MAST & DOWN-CONDUCTOR · MAST BASE · A PERSON · DIAGNOSIS

SCENE 37W

Four devices were on the outstation loop in August. EMC engineer Lauwers has each device's operating state and post-strike condition on separate cards. She wants one explanation that fits all four without changing the facts between cards.

GUIDE 60W

Four observations, and the two survivors are as informative as the casualties. The cable insulation still holds its test voltage. A battery logger three metres away is untouched. Ask of each candidate how many of the four it covers. What differs between the damaged pair and the logger is not the storm. It is how much area the circuit encloses.

ASK 6W

Which explanation fits all four observations?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a loop's induced voltage grows with the area it encloses

VERDICT 83W

The cable insulation still holds its test voltage, so a breakdown through the insulation does not fit. The battery logger three metres away is also unharmed despite sharing the storm environment. What differs is enclosed loop area. The long trench pair encloses substantial changing magnetic flux; the small battery logger does not. The trailer earth also measures the same before and after. Together, those observations point to inductive coupling through the trench geometry rather than damage delivered through the soil or failed insulation.

TAKEAWAY 16W

> A good failure explanation must fit both the damaged equipment and the nearby equipment that survived.

9.3 What the finding costs the plan

LAUNCH & RECORDS · LAUNCH CONTROL · A ROOM · CHOICE

SCENE 39W

Hal Brenner points at the trench drawing. If the coupling is geometric, every remaining shot is also a test of that geometry. Adeyinka Vero asks the harder question: does knowing the mechanism argue for firing more, or stopping sooner?

GUIDE 65W

Four readings of what the finding does to the remaining shots. Ask of each whether it treats the mechanism as an explanation or as a prediction. A formula can be computed before the next stroke and then compared with it. One option would remove the configuration the prediction needs. That is the real argument in the room, and it is not about who suspected what.

ASK 10W

What does the coupling finding change about the remaining shots?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a mechanism that is understood can be measured deliberately

VERDICT 81W

Before today, the outstation damage was an event the station could only survive. Now the mechanism has a formula and a numerical prediction. The next stroke can therefore be predicted before it is measured. If the result holds, the geometry becomes a design rule rather than a post-mortem explanation. Rerouting now would remove the very configuration needed to test that prediction. That is the real argument between Vero and Brenner: whether one more dangerous-looking configuration is also one more decisive measurement.

TAKEAWAY 19W

> Once a mechanism makes numerical predictions, the next event can test the mechanism rather than merely repeat the observation.

END OF THE DAY 16W

A circuit that touches nothing can still be driven, because flux does not need a connection.

DAY 10 OF 15

Nine volts and a hundred and eighty thousand

PLAN CARD 108W

Friday. Sifuentes has brought the April certificate back to the table. Yesterday's finding explains the trailer. It does not explain the rack at the mast base. That rack also lost two channels last August. It is bonded to the earth grid by six metres of heavy copper. Strand says six metres of copper is three tenths of a milliohm, and can be ignored. Sifuentes says that is the right number for the wrong quantity. One of them has to give. Today you work out what a fast rising current does along a straight piece of wire. Then you settle what the April certificate was ever able to say.

BRIEFING 17W

Six metres of copper between the mast base and the equipment earth, and nobody has costed it.

WHAT YOU DECIDE 13W

Work out what the bonding lead does, and what the certificate never covered.

10.1 The same wire, twice

EARTHING & TRANSIENTS · EARTHING COMPOUND · A ROOM · DERIVE

SCENE 40W

Six metres of 70 mm^2 copper run from the rack earth bar to the grid. Its measured resistance is 0.3 mΩ. Sifuentes places the April certificate beside the drawing while the strike's 1 μs rise time goes on the board.

GUIDE 60W

Build it a line at a time. The same six metres of copper has two voltages across it: one from its resistance and one from its inductance. Work out both — Ohm's law for the steady case, and the inductive voltage for the front of the strike. Then compare them, because the comparison is the answer rather than either number.

ASK 23W

Work out what the bonding lead does during the front of the strike, and compare it with what it does at direct current.

BEHIND THE BUTTON 164W

Why one lead has two answers. Resistance responds to how much current is flowing. Inductance responds to how fast it is changing. At direct current the second term is zero, so the certificate measured only the first. During a one-microsecond front, thirty kiloamps arrive at 3×10^{10} amps a second, and the inductive term is thousands of times larger.

Why the April certificate is not wrong. It measured what it claimed: 0.3 milliohms of resistance, correctly, at direct current. The error is in reading it as a statement about what the lead does during a strike. A certificate is only a claim about the conditions it was measured under.

What the number means for the rack. Kilovolts across a bonding lead appear between the rack earth and the grid, which is a difference no equipment inside was designed for. That is why lightning protection is about inductance and geometry rather than about conductor size, and why a fatter cable does not fix it.

TAKES AS READ

a straight conductor has an inductance of roughly a microhenry per metre · a changing current through an inductance produces a voltage across it

VERDICT 77W

The same six-metre lead gives only 9 V from IR at 30 kA, yet about 180 kV from L·dI/dt on the one-microsecond front. The wire did not change. The rate of current change did. Sifuentes looks back at the April certificate and says the important sentence herself: the 25-ohm result is correct. It measured the steady regime. A steady test makes the inductive term vanish, so the certificate could never contain the quantity that dominated the strike.

TAKEAWAY 15W

The same wire can be resistance-dominated at steady current and inductance-dominated during a fast transient.

10.2 Short, or not at all

FIELD & CHARGE · FIELD STATION · A PERSON · CHOICE

SCENE 39W

Vero stands at the rack with the six-metre bond marked on the drawing. The rack, mills and trailer each have their own connection sketches. She wants one site rule the crew can apply before the next installation is built.

GUIDE 63W

Four bonding rules. Ask of each what term it acts on. Cross-section moves resistance far more than it moves inductance, and inductance is what the fast front sees. Length and loop area are what inductance depends on. Two of the options would create differences between separate earths rather than remove them. The rule has to be usable before the next installation is built.

ASK 12W

What does the inductive term imply for how equipment should be bonded?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

inductance grows with the length of the conductor

VERDICT 80W

Inductance depends strongly on conductor length and on the loop made by its return path. Increasing cross-section changes resistance much more than it changes inductance. For a fast transient, shortening the bond is therefore more effective than simply making it thicker. Bringing equipment to one reference point also avoids large differences between separate earths. The practical rule is geometric: keep bonds short and keep the return path controlled. That directly attacks the voltage term that dominates during the strike front.

TAKEAWAY 15W

> Fast-transient bonding depends on geometry and shared reference paths, not only on low DC resistance.

10.3 Making a microsecond on purpose

IMPULSE HALL · A ROOM · CONTROL

SCENE 38W

Strand can fire the bank into the mast base with a front chosen from 100 microseconds down to about half a microsecond. Ruiz has one afternoon of hall time and asks which setting would actually test the claim.

GUIDE 58W

One afternoon, one variable. The front time is what separates the two explanations of the placard voltage, so change that and hold everything else — same bank, same electrode, same earth. Run it, and if the reading follows the front time both ways you have the answer; if you changed the electrode as well, you have a story.

ASK 14W

Which variable, changed on its own, separates the inductive explanation from the resistive one?

BEHIND THE BUTTON 137W

What the two explanations are. The voltage across the earth path is the resistive term plus the inductive one: IR while the current is steady, and L dI/dt while it is changing. A slow front makes the second term small, and a fast one makes it dominate.

Why the front time is the discriminating variable. Twenty-five ohms into thirty kiloamps is the same 750 kV whichever front you choose. What changes with the front is the inductive contribution, so a reading that rises as the front shortens is the inductive term and nothing else.

Why one variable at a time is the whole afternoon. Changing the front and the electrode together produces a number that is consistent with either explanation, and the hall is booked once. Reversing the change is what separates a cause from a coincidence.

TAKES AS READ

a laboratory impulse can be given a chosen rise time

VERDICT 189W

The placard voltage has two candidate sources and one afternoon to separate them. Across the earth path an inductive volt and a resistive volt add: IR + L dI/dt, where L over R sets the time the inductive part matters for. Impedance at a microsecond is not the same as resistance, and the resistive part does not care how fast the current arrives — 30 kA through 25 ohms is 750 kV at any front time. The inductive part cares about nothing else. So the front time is the one variable whose change moves one explanation and not the other, and the test is to move it alone: same bank charge, same electrode geometry, same earth connection. At a 100 microsecond front the measured peak sits near the resistive figure; at half a microsecond it rises to roughly 2.7 times it, and restoring the slow front brings it back down. That reversal is what makes it a finding rather than a coincidence that happened during the same

> afternoon. Change the electrode at the same time and the number is consistent with both models, and the hall is booked once.
>
> **TAKEAWAY** 15W
>
> A causal test changes the variable that separates competing models while holding confounding variables fixed.

END OF THE DAY 18W

A perfect earth at the end of six metres of wire is not an earth at a microsecond.

DAY 11 OF 15

Where the current goes

PLAN CARD 113W

Monday of week five. It is the quietest day the season has had. No cell is forecast before Wednesday. The hall is down for a gap service. Nakata has last week's stroke recorded at every shunt on the mast, which is a first. Tate has been asked to check the bonding at each joint while the weather still allows a climb. Today you read the current at each point down the conductor, and you find where the record stops adding up. The same records say how much charge the cell over the flat was holding. Nobody has added that up either. A quiet day is the only day this work gets done on.

BRIEFING 15W

Nothing goes wrong today. Three things get confirmed and one of them is a surprise.

WHAT YOU DECIDE 12W

Follow the strike down the mast and find where it stops matching.

11.1 Down the conductor, one station at a time

MAST & DOWN-CONDUCTOR · MAST BASE · A ROOM · PROBE

SCENE 60W

Thirty kiloamps entered the mast tip on the last stroke. Current that enters a conductor leaves it somewhere, so the same thirty should appear at every shunt down the conductor unless something is taking a share. Nakata has seven measuring points from tip to soil, deliberately not read in order, and wants the break located before anybody starts explaining it.

GUIDE 68W

Nothing is read for you. Press Read on any station and it reports what that station measured on the stroke, and what the station above it implies should have arrived. Take at least three. A shortfall between two stations is not an instrument error — it is a route the current took instead. Then click the station where the current stops being where it should be, and commit.

ASK 24W

Thirty kiloamps entered at the tip. Take the readings you think you need — where does the current stop being where it should be?

BEHIND THE BUTTON 185W

Why a shortfall is a route. Charge is conserved, so current does not go missing. If 29 kiloamps pass one shunt and 18 pass the next, then 11 left the conductor between them, and the only question is through what. On a mast that is usually a bond nobody drew on the diagram: a conduit, a cable tray, a ladder, an instrument earth.

What a shunt actually measures. Each station is a low-value resistor in the path with the voltage across it recorded, so the reading is the current through that station and nothing else. It cannot tell you where current went, only that it is no longer here — which is why the answer is a section between two stations rather than a component.

Why the order of reading does not matter and the pairs do. Every reading is of the same stroke, so they can be taken in any sequence. What carries information is the difference between two of them, which means a reading is worth little on its own and a pair either side of the suspect section is worth the trip.

TAKES AS READ

current entering a conductor leaves it somewhere · two conductors bonded together share whatever current is offered to them

VERDICT 81W

The upper shunts stay near 30 kA, including 29 kA at fifteen metres. The base then falls to 18 kA. That locates the missing current before any mechanism is chosen. About 12 kA leaves the down-conductor in the last fifteen metres. The conduit bond lies inside that interval. Nakata stops at the drawing because the implication is large. A path treated as instrumentation hardware has been carrying roughly a third of the strike current without appearing on the original current map.

TAKEAWAY 16W

Current is conserved, so a shortfall between two stations is not an error but a route.

11.2 Eight tenths of a coulomb

FIELD & CHARGE · FIELD STATION · A ROOM · BALLPARK

SCENE 30W

Halvorsen has the layer's charge density from day two and the cell's area off the radar. Vero wants the total in the season log, next to the strokes it produced.

GUIDE 59W

Five numbers, and three of them belong to other questions. The height of the base. The field at the ground. The charge in the August stroke. Ask of each whether this total depends on it. And note what comes out. It is less than one stroke moved, which is not a contradiction: charge separation continues while the storm runs.

ASK 11W

Estimate the charge held on the base of the cell overhead.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

charge per unit area multiplied by area is charge

VERDICT 88W

The estimate gives about 0.8 C on the charged cloud base, while the August stroke moved about 3 C. That is not a contradiction. A thundercloud is not a capacitor that discharges once and stops. Charge separation continues while the storm evolves, so the source can be replenished between strokes. The estimate is still useful because it connects a local field measurement with a total over the radar area. It also shows why one static cloud snapshot cannot be treated as the full charge budget of a flash.

TAKEAWAY 23W

A charge density is a local quantity and a total needs an area, which is the thing a single instrument can never supply.

11.3 Two hundred amps: buy the discriminating check

EARTHING & TRANSIENTS · EARTHING COMPOUND · A PERSON · VALUE

SCENE 39W

The trailer rod carried 200 A during a stroke that never touched it. Ana Sifuentes has the network map open beside the log. Four follow-up checks are possible before the next cell, but field access allows only two credits.

GUIDE 54W

Two credits, four checks, and one puzzle: the trailer rod carried 200 amps during a stroke that never touched it. Open each check and ask what result would separate ordinary soil conduction from a different path. Buy the two that can tell those apart, not the two that would measure the current more precisely.

ASK 19W

Which evidence is worth buying to decide whether the remote current is ordinary soil conduction or a different path?

BEHIND THE BUTTON 149W

What the two explanations are. Current spreading through soil from a distant strike falls off in a predictable way with distance, so 200 amps at the trailer would imply a whole pattern across the network. A different path — a buried service, a cable screen, a fence line — delivers current to one place without that pattern.

Why precision does not help. Measuring the 200 amps again to three figures leaves both explanations standing. The check worth buying is one whose two possible results point at different explanations, which is the same argument every VALUE stop makes.

Why the answer matters before the next cell. If it is soil conduction, the network map is right and the number is expected. If it is a path nobody drew, then equipment on that path saw current it was not designed for, and the same thing will happen in the next storm.

TAKES AS READ

soil conducts, and current spreads through whatever is available

VERDICT 85W

A small current at a remote rod does not identify its own path. An undocumented metal bond, a branch strike or a driven cable can all sound plausible from one number. A soil-potential map asks a different question: does the ground itself carry a continuous voltage gradient from the station toward the rod? If it does, distributed soil conduction explains the current without inventing a hidden connection. Repeating a nearby electrical check may add confidence, but it does not distinguish the competing routes as directly.

TAKEAWAY 17W

When several mechanisms fit one reading, the best follow-up is the measurement whose possible outcomes separate them.

11.4 What the round is doing in a storm

FIELD & CHARGE · FIELD STATION · A ROOM · SPOT

SCENE 32W

One standing rule governs work on the mast. The safety officer changes it as the storm comes in. The field mill rises, then a strike lands within ten kilometres, then the all-clear.

GUIDE 53W

Do the jobs the current rule allows and leave the rest. The rule is on the board at the mast base and it changes without an announcement. What is scored is the jobs either side of a change, because they are the only ones that show whether the crew is reading the board.

ASK 12W

Work to the rule on the board, and keep watching the board.

BEHIND THE BUTTON 97W

Why the rule changes with the field. A rising field mill means charge is building overhead and the mast is the tallest thing on a salt flat. The rule tightens in stages rather than all at once, because stopping all work at the first cloud would mean no work in a storm season.

Why the changeover is where somebody gets hurt. A crew working to the last rule is a crew on the mast after the rule said come down, and nothing about being three quarters of the way through a job makes the field any lower.

TAKES AS READ

work near a tall mast in a storm is governed by rules that change

VERDICT 143W

Three rules run across the storm and each allows a different part of the round. Anything at all, then ground-level work only, then nothing outside at all. A job can be allowed under two of them, which is what makes the change cost real rather than notional, and the panel scores the window either side of each change. Most of the list is allowed by neither rule or by both, and is handled correctly by somebody who has read nothing. The mast's own version of this is the two minutes after the field mill crosses its threshold: the rule has changed to ground level only, and there is somebody at forty metres who is nearly finished. Being nearly finished has never lowered a field, and the whole point of the staged rules is that they change before the strike rather than after it.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

END OF THE DAY 25W

A current that is not where it was a moment ago has gone somewhere, and the place it left is a fact about the site.

DAY 12 OF 15

Predict it, then measure it

PLAN CARD 109W

Tuesday. A cell is forecast for two o'clock. Vero wants the season's first prediction on the board before it arrives. Peak current, written down in advance, from the field and the charge the station has now measured. Brenner has the nowcast open and does not like how fast the cell is tracking. Today you commit to a number, then fire, then measure what came down. The afternoon does not end well. The crew comes in late off the flat. A trailer is left in a state nobody can use. That is the argument Brenner has been making since week one, and he was right in a way nobody wanted.

BRIEFING 16W

A cell for the afternoon, and a prediction that has to be written before it arrives.

WHAT YOU DECIDE 15W

Commit to a number before the stroke, and find out what a forecast is worth.

12.1 The number on the board

LAUNCH & RECORDS · LAUNCH CONTROL · A ROOM · VERIFY

SCENE 37W

The board has a space for a predicted peak current and Vero wants it filled before the rocket leaves the rail. The clamp will give the answer forty minutes later, if anybody goes out and reads it.

GUIDE 53W

Lock a predicted peak current before the rocket flies. It cannot be edited afterwards, which is the point. The rocket goes up whatever you write. Then decide whether to spend half an hour of a storm afternoon walking out to read the clamp. Not measuring is an answer too, and a poor one.

ASK 21W

Commit to a peak current for this stroke, fire the shot, and then decide whether to go out and read it.

BEHIND THE BUTTON 139W

Why the prediction is locked. A number written after the event is not a prediction. The failure is not dishonesty but memory: with the clamp reading in front of you, it is genuinely hard to recall how confident you were. Committing first is a procedure that protects you from yourself.

What the walk actually costs. Forty minutes, on an afternoon with a cell coming in, is time not spent on the mills or the launch rail. That is a real cost, and it is why the panel makes you decide rather than assuming somebody will go.

What is lost by not going. An unmeasured prediction can never be wrong, which is exactly what makes it worthless. A station that predicts and never checks accumulates confidence rather than calibration, and the review has no way to tell the two apart.

TAKES AS READ

a prediction is only a prediction if it is recorded before the outcome · reading an instrument after the event is a decision with a cost

VERDICT 82W

Prediction and measurement are separate acts, and only the pair tests the model. A number written before the shot can be wrong. A number never checked cannot be. The accepted range is deliberately broad because the field reading predicts only the scale of the peak, not an exact current. Going to the clamp afterwards is therefore part of the experiment, not paperwork. The station learns only when the committed prediction is compared with a measurement that had a real chance to disagree.

TAKEAWAY 18W

Predicting and measuring are two separate acts, and skipping the second one is its own kind of failure.

12.2 The afternoon Brenner is right about

COUPLING & THE OUTSTATION · OUTSTATION · A ROOM · CHOICE

SCENE 37W

At 2:00 the second cell is thirty kilometres out. By 2:20 it is eleven kilometres away. The crew door closes at 2:25. At 2:30 the first natural strike flashes and the outstation telemetry disappears from the wall.

GUIDE 60W

Four candidates for what failed, and everybody did what they were told. Ask of each whether the evidence supports it. The rule was followed. The crew were inside. The second cell was tracked the whole way. What was left was five minutes, where the projection had allowed more. Something in the plan was wrong, and it was not the people.

ASK 4W

What failed this afternoon?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a nowcast projects a cell forward from where the radar last saw it

VERDICT 83W

The crew followed the rule and still lost the margin. The second cell covered nineteen kilometres in twenty minutes, faster than the projection had allowed. Everyone was inside before the first natural strike, but only by five minutes. That is Brenner's argument becoming an observation. The problem is not that Vero ignored the rule or that the crew moved slowly. The lead-time model itself failed on this cell. Day four showed the same rule working; today shows the case it did not cover.

TAKEAWAY 15W

A lead-time rule must survive real variation in the process it is trying to outrun.

12.3 What survives a shorter season

IMPULSE HALL · A PERSON · STRESS

SCENE 39W

A strike has taken the outstation trailer out for the season. Nobody was inside. The mast, shunts and grid still work, while the impulse hall remains available. Vero and Brenner need a plan before the next storm window opens.

GUIDE 61W

Slide the usable storm days down to five and watch which plans go dark. A plan has to keep safe field measurements possible and fit inside the season that is left. The trailer is gone for the year, so anything that depends on it is not a plan. Choose the one that still stands at the pessimistic end of the range.

ASK 18W

Which remaining-season plan still works if the station gets far fewer usable storm days than the calendar suggests?

BEHIND THE BUTTON 142W

Why the pessimistic end rather than the middle. The number of usable storm days is a forecast with a wide spread, and a plan chosen at the expected value fails half the time. Testing at the low end costs nothing now and is the only version of the choice that survives a poor season.

What the strike actually removed. The outstation trailer is out and nobody was inside it. So the loss is instrumentation coverage rather than people, and the plans that leaned on remote measurements have lost their instrument. The mast, the shunts and the grid still work.

Why the impulse hall changes the answer. Bench work does not need storm days at all. A plan that shifts weight onto the hall is a plan whose progress does not depend on the weather, which is exactly what a shortened season rewards.

TAKES AS READ

a station has a fixed amount of time and more than one way to spend it

VERDICT 85W

Rebuilding the trailer can recover the original measurement set, but it spends storm days the station is trying to use. Hall-only work is safe but gives up the remaining natural-strike record. Continuing unchanged ignores the lead-time failure that just occurred. The mixed plan keeps the mast programme, moves coupling tests into the hall and tightens the field call-in rule. It is not the richest plan under a generous season. It is the only one that still produces safe field evidence when usable sky becomes scarce.

TAKEAWAY 20W

A robust plan is one that still meets its essential constraints when the scarce resource arrives at the pessimistic end.

END OF THE DAY 16W

A prediction that is never measured against anything is an opinion with a unit after it.

DAY 13 OF 15

What the mill cannot see

BEFORE THE DAY — CANVASS

Who else saw the strike land west of the mast

Something struck west of the station at four and only one channel recorded it. Ask round the crew and the two neighbouring farms until enough accounts agree on where it landed, because a single-channel record is not a measurement of anything yet.

PLAN CARD 117W

Wednesday. The record of Tuesday's stroke has a hole in the middle of it. The clamp caught the current. The shunts caught the current. All four field mills recorded a smooth dip instead. That dip only begins after the stroke was already over. Mira Halvorsen has said for two seasons that the mills are slow. Nobody has ever asked her how slow. Kaya needs a caption for the plot before the pack closes. Today you solve the electrometer circuit inside the mill and get its time constant out of that solution. A number off a data sheet will not do here. Then you say what the mill record is evidence about, and what it is silent on.

BRIEFING 13W

The mills report the slow field, and nobody has written down how slow.

WHAT YOU DECIDE 12W

Solve the electrometer's circuit, and find out what its time constant costs.

13.1 The circuit that decides what is recordable

EARTHING & TRANSIENTS · EARTHING COMPOUND · A ROOM · DERIVE

SCENE 37W

The field mill's input is 10 MΩ across about 100 pF. Halvorsen puts the two numbers beside a one-microsecond stroke front. The question is whether the instrument can follow that event or only what remains after it.

GUIDE 54W

Build it a line at a time. The loop equation gives a differential equation in the charge, and the current out of the capacitor is minus dq/dt. Separate and integrate to get the charge as a function of time, then read off the time constant. Finish by comparing it with a one-microsecond stroke front.

ASK 12W

Solve the electrometer's discharge, and get the time it needs to respond.

BEHIND THE BUTTON 142W

Why this is the same equation as the river. Charge leaving in proportion to how much is there gives an exponential decay, which is the recession curve, the cooling cup and the decaying nucleus. Recognising the form is most of the work, and the constant that comes out has units of time.

What the time constant tells you. R times C is the time to fall to about a third. Ten megohms across a hundred picofarads gives a millisecond, which is a thousand times longer than the microsecond front of a stroke.

What that means for the instrument. The electrometer cannot follow the front at all; it reports what remains once the fast part is over. So the mill's record is a statement about the slowly varying field, and reading a stroke's peak off it is reading something the circuit never saw.

TAKES AS READ

the loop rule holds around any closed circuit · the current out of a capacitor is the rate at which its charge falls

VERDICT 82W

The input time constant is $RC = 10\ \mathrm{M\Omega} \times 100\ \mathrm{pF} = 1\ \mathrm{ms}$. A stroke front lasts about $1\ \mu\mathrm{s}$, so the instrument responds a thousand times too slowly to follow it. The mill is not broken and the data are not noise. The circuit averages changes that are much faster than its response. That makes the post-stroke field shift useful evidence about cloud charge, while the microsecond front is effectively absent. Bandwidth decides what an instrument can honestly claim.

TAKEAWAY 23W

The exponential is not assumed; it is what comes out of the loop rule when the current is the derivative of the charge.

13.2 The dip after the stroke

FIELD & CHARGE · FIELD STATION · A PERSON · CHOICE

SCENE 36W

All four mills show the same thing: no feature at all during the stroke, then a smooth recovery over the following few seconds. Kaya has to write a caption for the plot in the season report.

GUIDE 57W

Four captions for the same plot. Ask of each which timescale it claims. The mill responds in milliseconds, so a microsecond front was averaged away before anything recorded it. What the mill does resolve is the step before and against after, which lasts seconds. Calling the instrument useless is as wrong as claiming it saw the front.

ASK 10W

What is the mill record of Tuesday's stroke evidence about?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an instrument reports an average over its own response time

VERDICT 81W

The field step before and after the stroke persists much longer than the mill's millisecond response. That makes it a real measurement of the source changing. The microsecond front is different. The circuit averages it away before the recorder can follow it, and later processing cannot reconstruct information the instrument never resolved. Calling the mill useless would also be wrong. It measures the slower electric-field change well. The record needs a caption that states which time scale the instrument actually observed.

TAKEAWAY 22W

> A slow instrument is evidence about slow things, and its silence during a fast event is not a measurement of the event.

13.3 What to put on a microsecond

MAST & DOWN-CONDUCTOR · MAST BASE · A ROOM · ALLOCATE

SCENE 31W

Nakata has four instruments in the cabinet and a request from the review for the rise time of the next stroke. Each one has a response time written on its case.

GUIDE 59W

A fixed afternoon of installation time, and a list of instruments that wants more of it than there is. Buy the ones that can actually see a microsecond front, and read off what your choice stops the station being able to report. Everything here measures something real; what the plan cannot answer is the price of what it can.

ASK 17W

One afternoon of installation time. Fit the channels, and know what the fit stops the station reporting.

BEHIND THE BUTTON 149W

What a response time does. An instrument reports honestly inside its own band and produces a number outside it — a smoothed, late, smaller version of what happened. That number is wrong rather than merely imprecise, and nothing in the record says so.

Why a microsecond front is the hard case. To resolve a rise you need a response several times faster than the rise itself. A 20-millisecond instrument facing a 2-microsecond front reports a broad low bump: the peak is understated by a factor of ten and the front is invisible.

What the slow instruments are still for. Charge transferred, continuing current and the hour-scale field are all real quantities that the fast channels do not capture, and the review is not the only reader of this record. The plan is a choice about which questions the station can answer next season, not about which instruments are good.

TAKES AS READ

an instrument reports honestly only within its own band

VERDICT 150W

An instrument's response time decides which questions it can answer, and a front of about two microseconds needs a channel several times faster than that. The 10 nanosecond current transformer and the 100 nanosecond field mill can both resolve it. The 20 millisecond electrometer and the hour-scale field logger cannot, and they will not fail visibly — they will report a smooth low bump with the peak understated by a factor of ten and no flag anywhere in the record. That is what makes an out-of-band measurement worse than a missing one. But the slow channels measure real quantities the fast ones cannot: charge transferred, continuing current, the diurnal field. The afternoon does not stretch to everything, so the plan is a decision about which questions the station can answer next season — and the honest version of it says out loud that total charge is not one of them.

TAKEAWAY 18W

Choosing an instrument is choosing a band, and a measurement outside it is wrong rather than merely imprecise.

13.4 A strike, or the instrument

LAUNCH & RECORDS · LAUNCH CONTROL · A ROOM · BELT

SCENE 40W

The night produced more triggers than the log can carry. Some of them are current going down the mast. The rest are the recording chain reacting to the same storm, and both arrive as a trace with a rise time.

GUIDE 53W

Two bins. A real stroke arrives on every channel that should see it, with the timing the geometry allows, and with a rise the instrument can actually resolve. A chain artifact appears on one channel, or arrives before the trigger, or has a rise faster than any sensor on the site can report.

ASK 16W

Send each trigger to the bin that says whether the storm or the chain made it.

BEHIND THE BUTTON 80W

Why bandwidth is the test. An instrument cannot report a rise faster than its own response time, so a trace that rises faster than the sensor allows was made after the sensor — in the cable, the digitiser or the trigger.

Why coincidence matters. A strike is one event and the site has several instruments. If the field mill, the current transformer and the two remote stations disagree about whether anything happened, the thing that happened was in one box.

TAKES AS READ

an instrument can only report what it is fast enough to see

VERDICT 140W

Everything in this log has been through a sensor, a cable and a digitiser. Each of those can make a trace of its own. Two tests separate them: bandwidth and coincidence. No instrument can report a rise faster than its own response time. So a front that rises in twenty nanoseconds, on a channel whose sensor takes a microsecond, was made after the sensor — in the cable, the amplifier or the trigger. And a strike is one event seen by a site full of instruments. If the field mill, the current transformer and the two remote stations do not agree that something happened, then what happened was inside one box. So an honest log records the coincidence and the rise time rather than the peak. It is also why a night of four hundred triggers can hold nine strikes.

TAKEAWAY 12W

A trace faster than the instrument was not made by the weather.

END OF THE DAY 24W

An instrument's time constant decides what it is evidence about, and no amount of care with a slow instrument makes it a fast one.

DAY 14 OF 15

The last thing that can be changed

PLAN CARD 114W

Thursday. This is the last day anything can be altered. The final impulse test runs tomorrow, at whatever setting is agreed this afternoon. One gap service is left. The review pack closes on whatever that shot returns. Strand wants twelve stages at fifty kilovolts. That is what the hall has fired eleven hundred times. Lauwers wants the same peak with a shorter front instead. A short front tests Tuesday's coupling number. The long one confirms the placard. Today you work out what twelve stages in series actually produce, and you choose what the last shot of the season is for. The report also has to recommend something to a station nobody here has visited.

BRIEFING 13W

After today the hall is committed and the sky is whatever it is.

WHAT YOU DECIDE 14W

Set the bank for the season's final test, and say what it will show.

14.1 In parallel to charge, in series to fire

IMPULSE HALL · A ROOM · DERIVE

SCENE 35W

Twelve stages, each a hundred nanofarads, all charged side by side to fifty kilovolts. When the gaps break down they are in a chain instead. Ruiz wants the output written before the setting is agreed.

GUIDE 59W

Build it a line at a time. The stages charge side by side and fire in a chain, so the charging voltage and the output voltage are different questions. Work out what the chain delivers, then the series capacitance, then check the stored energy against the parallel case. The energy has to agree, and that check is the point.

ASK 16W

Work out what the bank delivers when the gaps fire, and check it against the energy.

BEHIND THE BUTTON 144W

Why parallel to charge and series to fire. In parallel every stage sees the same supply, so a modest supply can charge all twelve. When the gaps break down the stages are connected end to end, so their voltages add. That is a Marx generator, and it is how a fifty-kilovolt supply produces six hundred kilovolts.

What happens to the capacitance. Capacitors in series combine reciprocally, so twelve hundred-nanofarad stages in a chain behave as one stage of about eight nanofarads. Voltage up by twelve, capacitance down by twelve.

Why the energy check is the real answer. Half C V squared with the series values gives the same total as twelve individual stages did while charging, because nothing has been created. Any answer where the energy grows has an arithmetic error in it, and the check will find it before the setting is agreed.

TAKES AS READ

capacitors in series carry the same charge and their voltages add · the reciprocals of series capacitances add

VERDICT 74W

When the twelve stages fire in series, the bank produces about 600 kV and an effective capacitance near 8.3 nF. Those values still store the same 1,500 J present before firing. The switching network does not create energy. It trades capacitance for voltage by the stage count. That is the key check on the derivation. A Marx generator can therefore reach hundreds of kilovolts even though no individual charging supply ever exceeds 50 kV.

TAKEAWAY 24W

Switching from parallel to series multiplies the voltage by the number of stages and divides the capacitance by it, and the energy does neither.

14.2 Repeat the placard, or test the number

LAUNCH & RECORDS · LAUNCH CONTROL · A PERSON · VALUE

SCENE 37W

One shot left in the hall and two proposals for it. Strand's runs the standard setting and confirms the bank. Lauwers's halves the front at the same peak and predicts a specific voltage on the bench loop.

GUIDE 53W

One shot, and a list of things it could be spent on that costs more than one shot. Buy what would change the station's next decision rather than what would be reassuring to have. Two of these confirm something already known to two significant figures, which is worth exactly nothing to the review.

ASK 11W

One shot in the hall. What is it worth spending on?

BEHIND THE BUTTON 138W

What makes a test worth firing. Not how precise it is and not how likely it is to succeed — whether its possible outcomes point to different decisions. A shot whose two outcomes both leave the placard where it is has told the station nothing it can act on.

Why a prediction with a number is the strong form. Lauwers's proposal names the voltage the bench loop should reach if the inductive account is right. It can come out wrong, and a test that cannot come out wrong is a demonstration.

What the repeat is actually for. Confirming the bank at the standard setting is defensible when the bank is the thing in doubt. It is not, and it has been fired at that setting eleven times this season, so the eleventh confirmation buys the same axis again.

TAKES AS READ

a test is worth what it could change

VERDICT 177W

A test earns its shot by changing a decision. Lauwers's proposal halves the front at the same peak current and names the voltage the bench loop should reach if the inductive account of the placard number is right — a prediction that can fail, which is what makes the shot informative either way. If the bench reads what she says, the placard is revised on the inductive term. If it does not, the resistive account survives and the mast keeps its current figure with a documented reason. Strand's repeat at the standard setting confirms a bank that has already been fired there eleven times this season, and the twelfth confirmation is the same axis again. The two calibration items are real work and belong in a maintenance window, not in the last shot of an afternoon: they improve numbers nobody's next decision turns on. And the shot itself is irreversible — the hall is booked once, so the choice is not which test is best in general but which one this station cannot answer any other way.

TAKEAWAY 15W

> The most informative test is the one whose possible outcomes can falsify an important model.

14.3 What to tell the next station

COUPLING & THE OUTSTATION · OUTSTATION · A ROOM · CHOICE

SCENE 30W

Lauwers has to write a paragraph the review will read as a recommendation. Three mitigations are on the table and only one of them is supported by this season's measurements.

GUIDE 57W

Four mitigations, and all four are sensible engineering. Ask of each whether this season measured what it buys. The coupling depends on the length of the parallel run and on how close it gets to the conductor. Better earthing acts on a different path. And the trailer was already distant. Forty metres of its cable was not.

ASK 11W

What should the report recommend for a site like this one?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a recommendation is only as good as the measurement behind it

VERDICT 82W

The measured coupling depends on the length of the parallel run and the logarithm of its near and far distances from the conductor. Routing signal pairs away from that geometry therefore follows directly from the season's evidence. Better earthing may be useful, but it addresses a different path. Surge protection is sensible practice, yet this campaign did not measure how much it buys. Moving the trailer alone also misses the mechanism. The trailer was distant; forty metres of its cable was not.

TAKEAWAY 16W

A recommendation should target a variable that the season's evidence shows actually controls the measured mechanism.

END OF THE DAY 20W

A test that repeats what is already known spends the same time as one that could come out either way.

DAY 15 OF 15

What the season can say

PLAN CARD 107W

Friday. The pack goes tonight. Vero wants the second mast. Brenner wants a lead time that comes from how fast cells have actually crossed this flat, not from the handbook. Sifuentes wants the April certificate reissued with the quantity it measured written on its face. All three go in or none of them does. The review reads the pack before it reads anybody in this room. Today you set the criterion the station will work to next season. You also sort the season's claims by what stands behind each one, and name the thing fifteen days of work has not shown. Nothing can be added after tonight.

BRIEFING 14W

The review pack closes tonight, and every sentence in it has to be answerable.

WHAT YOU DECIDE 13W

Set next season's rule, and sort what was measured from what was inferred.

15.1 Earlier, or more certain

LAUNCH & RECORDS · LAUNCH CONTROL · A ROOM · CLOUD

SCENE 33W

Twenty-two minutes is the lead time the station actually achieved this season, against the thirty the rule assumes. Four changes are available and each of them costs something the station has little of.

GUIDE 69W

The cloud is the lead time this station actually achieved across the season — twenty-two minutes on average against the thirty the rule assumes. The band is what the rule needs. Four changes are available, each costing something the station has little of, and they act differently: some move the average, some make the season more consistent. Apply what you would buy, then say when next season is defensible.

ASK 27W

Bring next season's lead time inside what the rule needs, and say when it is defensible. Place the bars to report the mean and its one-sigma uncertainty.

BEHIND THE BUTTON 172W

Why an average is not the claim. The rule is about every strike, not the mean one: a season that averages thirty minutes with a wide spread still contains days when the crew had ten. The cloud is the distribution of what actually happened, and the criterion is written on the fraction of it that clears the rule.

Moving and tightening. Calling the clear a mill-step earlier, or moving the shelter closer, adds minutes to every day — the cloud slides. A second radar feed with a nowcast, or a written and drilled call-out sequence, removes the days that went badly — the cloud tightens. Both buy lead time; only one of them removes the tail the rule is about.

What the tail costs. The days in the low tail are the days with people still on the flat when the first strike lands, and they are the days that end up in an incident report. That is why a plan is judged on its worst days rather than its typical one.

TAKES AS READ

a rule has to work across every administration of it, not on a good afternoon · lead time varies from cell to cell as well as having an average

VERDICT 82W

Calling earlier shifts the center of the lead-time distribution but does not make one afternoon resemble another. Shortening the walk does the same. The rule must survive the low end of the distribution, not only its average. A second radar feed reduces uncertainty in cell motion. A drilled call-out sequence reduces variation between shifts. Those changes narrow the spread itself. The final criterion is defensible only when both the center and the spread keep the required walk time inside the accepted band.

TAKEAWAY 19W

A rule fails on its spread as often as on its average, and only some changes touch the spread.

15.2 What the report can actually stand behind

EARTHING & TRANSIENTS · EARTHING COMPOUND · A ROOM · ATTEST

SCENE 40W

The funding-review copy closes tonight. Four headline claims are already written and signed. Two have direct backing, while two depend on steps the station has not physically verified. Kaya has time to trace only two claims before the report locks.

GUIDE 58W

Four headline claims, all signed, and two checks before the report locks tonight. Open each claim and read what already backs it. Two rest on direct evidence and two rest on steps nobody has physically verified. Hold what the backing does not support, and spend the checks where a wrong claim in a funding review would cost most.

ASK 14W

Which claims deserve the final two verification checks before the season report is frozen?

BEHIND THE BUTTON 126W

What a signature is worth here. It records that somebody took responsibility for a sentence. It does not record that the condition was observed. A claim can be signed, believed and true, and still have nothing behind it that a reviewer could inspect.

Why the funding review changes the stakes. This copy goes to people deciding whether the station continues. A claim that turns out to be unbacked costs more than one that was never made, because it puts every other claim in the document in doubt.

How to choose between two unbacked claims. Not by which is least likely — by which does more damage if it is wrong. That is consequence rather than probability, and it is the judgement the two checks are for.

TAKES AS READ

claims of different kinds are defended in different ways

VERDICT 81W

The 200 kV trench result has a derivation and a hall test behind it. The missing 12 kA is supported by measurements along the conductor. The 180 kV bonding-lead voltage is different: it was computed from length and dI/dt but never read directly. The routing recommendation then extends that mechanism to another station. Those last two claims carry consequence without equivalent backing. A final review should spend scarce verification effort there, rather than making the best-supported lines look even more complete.

TAKEAWAY 15W

A signed claim is not stronger than the weakest unverified step needed to support it.

15.3 What is still open

MAST & DOWN-CONDUCTOR · MAST BASE · A PERSON · CHOICE

SCENE 36W

The funding-review copy locks tonight. Nakata has one line left under “Unresolved.” Four candidates are on the screen. Kaya wants the line that names a physical claim the season still cannot support with a direct measurement.

GUIDE 61W

Four candidates for the unresolved line. Ask of each whether it names a physical claim that a measurement could settle. One of them is scheduled work. One is a design preference about instruments that are deliberately slow. One is a charge budget that does not control the mitigation. The season computed a voltage it never measured, and the copy locks tonight.

ASK 11W

What should the pack name as the season's main unresolved question?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a season produces limits as well as results

VERDICT 85W

The largest open claim is whether the predicted fast voltage actually appears across the bonding lead during a stroke. The season computed about 180 kV from its length and dI/dt, but never measured that voltage directly. A fast differential probe and suitable digitiser would test it. The other listed items are different. The mills are intentionally slow, the cell charge budget does not control the mitigation, and rebuilding the trailer is scheduled work. The report should name the unmeasured physical claim, not a task list.

TAKEAWAY 22W

> Naming what is unresolved is what makes the rest of a report usable, and it is the sentence a review reads first.

END OF THE DAY 18W

A season is worth what its record can defend, which is always less than what the room believes.

THE CLOSING CARD

The review funded the second mast, on the strength of one paragraph: the outstation was destroyed by two hundred kilovolts induced around a loop that touched nothing, computed from a length and a logarithm and then put at risk in the hall at a shortened front, where the bench loop read within nine per cent of the prediction. The April earthing certificate was reissued with the words "steady-state resistance" on its face. Next season starts on a lead time derived from how fast cells actually crossed this flat, which is twenty-two minutes rather than thirty, and on a call-out sequence somebody has to drill.

What it cost: a trailer, four working days, and an afternoon in week five when the crew came in late and Brenner was right in a way nobody wanted to be right. What is unfinished: the inductive volts along the bonding lead are still computed rather than measured, because the station owns no probe fast enough to believe; the conduit that carries a third of every strike into the instrument cabinet has been bonded, not rerouted; and nobody has yet fired a shot at a front short enough to test what happens when the lead is halved.

The second mast is yours. You computed the volts induced around a loop that touched nothing, you went and tested that number in the hall instead of believing it, and you put a lead time on next season that came from how fast the storms actually cross this flat. A station stayed open because one paragraph could be defended line by line. You wrote it.

Card text only — no options, boards or answers. 11,926 words of card prose across 52 stops. Generated from `themes/groundtruth` by `tools/make-cardbook.mjs`.